



DEPARTMENT OF ENGINEERING MATHEMATICS

Length Generalisation in Transformer-Based Sequence Models: Failure Taxonomy and Mitigation Strategies

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Declaration

This dissertation is submitted to the University of Bristol in accordance with the requirements of the degree of Master in Science in the Faculty of Engineering. It has not been submitted for any other degree or diploma of any examining body. Except where specifically acknowledged, it is all the work of the Author.

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Supporting Technologies

- Python was used to implement dataset generation, model training, evaluation, diagnostic analysis, and result visualisation.
- PyTorch was used to implement the Transformer models, positional encoding variants, optimisation procedure, checkpointing, and GPU execution.
- NumPy and pandas were used for numerical processing, aggregation of experimental results, and construction of reproducible summary tables.
- Matplotlib was used to produce the dissertation figures. A common visual style, colour mapping, typography, and output format were applied across the final plots.
- PyYAML configuration files were used to define experiment settings separately from the training and evaluation implementation.
- The aeon dataset tools were used to obtain and prepare the publicly available SelfRegulationSCP2 time-series dataset.
- Amazon Web Services EC2 G5-family GPU resources were used for the multi-seed training and evaluation runs that were impractical to execute locally.
- Git and GitHub were used for version control and for synchronising the experimental implementation between local and AWS environments.
- LaTeX was used to prepare the dissertation and to include publication-quality PDF figures.

Notation and Acronyms

Acronyms

- ALiBi** Attention with Linear Biases.
- AWS** Amazon Web Services.
- EM** Exact match.
- FFN** Feed-forward network.
- GPU** Graphics Processing Unit.
- ID** In-distribution evaluation.
- LSTM** Long Short-Term Memory.
- NoPE** No explicit positional encoding.
- OOD** Out-of-distribution evaluation.
- PE** Positional encoding.
- RNN** Recurrent neural network.
- RoPE** Rotary Position Embedding.
- SD** Standard deviation.
- UEA** University of East Anglia.

Mathematical Notation

- n Sequence length.
- d_{model} Dimensionality of the Transformer representations.
- d_k Dimensionality of the key and query vectors within an attention head.
- N_{head} Number of attention heads.
- $\mathbf{X}$ Matrix of input representations supplied to a self-attention layer.
- $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ Query, key, and value matrices used in self-attention.
- $\mathcal{L}_{\text{train}}$ Set of sequence lengths represented during training.
- L_{train} Sequence length used to train a fixed-length configuration.
- L_{test} Sequence length used during evaluation.
- L_{max} Maximum sequence length represented during training, $\max \mathcal{L}_{\text{train}}$.
- $A_{\text{EM}}(L)$ Exact-match accuracy at test length L .
- $A_{\text{tok}}(L)$ Token accuracy at test length L .

$E_{\text{EM}}(L)$ Exact-match error percentage, $100 - A_{\text{EM}}(L)$.
 $E_{\text{tok}}(L)$ Token-error percentage, $100 - A_{\text{tok}}(L)$.
 $\mathcal{V}_{\text{target}}$ Set of possible target-output symbols.
 $|\mathcal{V}_{\text{target}}|$ Number of possible target-output symbols.
 E_{chance} Uniform-prediction token-error reference, $100 (1 - 1/|\mathcal{V}_{\text{target}}|)$.

Reproducibility Statement

The experimental implementation separates task generation, model definitions, training, evaluation, configuration files, raw results, and derived plots. Synthetic experiments were repeated using random seeds 42, 123, and 2024. Configuration files record the model architecture, positional encoding, training length, evaluation lengths, optimiser settings, and output locations. Final figures were generated from saved CSV results rather than manually entered values. Representative position-wise diagnostics use the seed-42 checkpoints and are explicitly distinguished from the multi-seed baseline analysis.

Abstract

Transformer models have become a dominant architecture for sequence modelling, partly because self-attention allows them to represent dependencies across an input without the recurrence used in earlier sequence models. However, strong performance on the sequence lengths observed during training does not necessarily imply that the same model has learned a rule that generalises to longer sequences. This limitation is important because many practical sequence-modelling problems require models to operate reliably outside the exact length distribution seen during training.

This dissertation investigates length generalisation in Transformer-based sequence models using a controlled train-short/test-long experimental framework. The project focuses on three synthetic sequence tasks: addition, delayed copy, and reverse. These tasks were selected because they test different forms of sequence reasoning, including symbolic computation, long-range memory, and position-dependent transformations. A compact Transformer model was evaluated across five positional encoding settings: learned positional embeddings, sinusoidal positional encodings, no explicit positional encoding, ALiBi, and RoPE. Experiments were repeated across three random seeds to assess robustness rather than relying on single-run results.

A baseline-eligibility procedure first identified configurations that reliably learned the training task. Eligible configurations were then analysed using exact-match accuracy, token-error percentages, position-wise error profiles, attention diagnostics, and task-specific tests of carry conditions and dependency distance. Mixed-length training, curriculum learning, and randomised padding were evaluated against matched original baselines to test specific hypotheses about exposure to varied lengths and dependence on absolute positions.

All five eligible task-encoding configurations that met the multi-seed baseline-eligibility criterion fell below the 10% exact-match threshold at the first unseen length. At this point, Addition showed approximately 86.9–88.0% token error, with errors increasing modestly towards later output positions. Delayed Copy and Reverse showed approximately 98.8–99.0% token error that was almost uniform across output positions. Addition errors differed by less than one percentage point between digits with and without an incoming carry for the learned model and were effectively identical for the sinusoidal model. Reverse errors were not concentrated at a particular dependency distance. Mixed-length variants underfit in the evaluated settings, while curriculum and randomised-padding variants that retained training-length performance produced no observed improvement at unseen lengths. In the limited SelfRegulationSCP2 extension, mean classification accuracy remained approximately 49–56% across the evaluated input lengths rather than showing the catastrophic exact-match collapse observed on the synthetic tasks.

The resulting failure taxonomy distinguishes training-length underfitting, seed instability, immediate extrapolation collapse, position-dependent error structure, and no observed improvement under tested mitigation. These findings show that length-generalisation evaluation should establish a valid training baseline before examining not only whether performance decreases, but how failure is distributed and which intervention hypothesis was actually tested.

Keywords: Length generalisation; Transformers; positional encoding; synthetic sequence tasks; extrapolation; failure taxonomy; baseline eligibility; position-wise diagnostics; mitigation strategies; time-series classification.

Chapter 1

Introduction

1.1 Context and Motivation

Sequence models are used in applications where the amount of available context can vary substantially between examples. A model may be trained using sequences of a particular length because of data availability, memory constraints, or computational cost, but later be required to process longer inputs. Reliable performance therefore requires more than fitting the length distribution represented in the training data: the model must also transfer the learned computation to sequence lengths that were not observed during training.

Transformers have become a widely used architecture for sequence modelling because self-attention allows direct interaction between positions without the recurrent computation used in earlier sequence models [22]. However, self-attention does not by itself guarantee that a model will learn a length-independent rule. A Transformer can achieve high accuracy within the training distribution while relying on patterns that are specific to the positions or sequence lengths encountered during training. Consequently, strong in-distribution performance may conceal a substantial failure when the same model is evaluated on longer sequences.

Length generalisation refers to the ability of a model trained on shorter instances to perform correctly on longer instances of the same underlying task. It is a form of out-of-distribution generalisation because the task rule remains unchanged while the sequence length moves beyond the training distribution. Previous studies have shown that this remains difficult for Transformer models, including on tasks for which a systematic algorithm should, in principle, apply at arbitrary lengths [1, 23]. This distinction is especially important for arithmetic and algorithmic tasks: success on short examples does not necessarily demonstrate that the model has learned the general procedure needed to solve longer examples.

Positional encoding is closely connected to this problem because a Transformer must represent token order while operating at positions that may not have appeared during training. Learned absolute embeddings, fixed sinusoidal encodings, rotary position embedding (RoPE) [19], ALiBi [15], and models without explicit positional encodings (NoPE) [9] make different assumptions about how position should be represented. Randomised positional encodings have specifically been proposed to expose models during training to positions beyond the ordinary range [17]. Nevertheless, published results vary across architectures, tasks, training procedures, and evaluation metrics. A positional encoding that assists one experimental setting cannot therefore be assumed to provide length generalisation universally. This problem is related to, but distinct from, extending the usable context window or benchmarking efficient long-context architectures [6, 21].

A further difficulty concerns how failure is evaluated. Reporting only accuracy at the longest test length does not establish whether the model learned the original training task, where extrapolation first deteriorated, or how errors are distributed within the output. For example, a configuration that never learns the training length represents training underfitting rather than length-extrapolation failure. Similarly, two models with zero sequence-level exact-match accuracy may exhibit different token-error percentages and different positional error structures. These behaviours require different interpretations even though their final exact-match scores are identical.

This dissertation addresses these limitations through a controlled train-short/test-long investigation of compact Transformer models. Addition, delayed copy, and reverse tasks are used to represent symbolic computation, long-range retention, and position-dependent transformation. The experiments compare five positional encoding settings across three random seeds, first establishing which configurations learn

the training task reliably and then examining how eligible configurations behave at progressively longer sequence lengths. The study combines exact-match and token-level metrics with position-wise, attention-based, and task-specific diagnostics. It also evaluates targeted training interventions against matched original baselines and includes a limited real-world time-series extension. The inclusion of this extension is motivated by the use of specialised Transformer architectures for long and irregularly sampled time-series data [13].

1.2 Research Problem, Aim, and Questions

Existing studies demonstrate that Transformer length generalisation is affected by positional representation, training procedure, task structure, and output format [1, 9, 17]. However, identifying which configuration achieves the highest accuracy does not fully characterise the problem. A model may fail because it never learned the training task, because its performance depends strongly on random initialisation, or because a solution that works at the training length does not transfer to longer sequences. These cases can produce similarly poor long-length accuracy while representing different experimental behaviours.

The central problem addressed in this dissertation is therefore not simply whether Transformer performance decreases as sequence length increases. It is how to distinguish the observable forms of failure, identify where errors occur within the predicted output, and determine whether an intervention changes the relevant behaviour when compared with an appropriate baseline.

In this dissertation, a failure taxonomy means a structured classification of empirically observed failure behaviours. The purpose of the taxonomy is not to assign a single label to every unsuccessful model, but to separate distinct experimental outcomes that have different interpretations, such as underfitting, extrapolation collapse, seed instability, and failure to improve under mitigation.

Importantly, negative results are treated as diagnostic evidence rather than as unsuccessful experiments. A poor result is interpreted according to whether the model fails to learn the training task, learns the training task but collapses beyond it, varies substantially across random seeds, or remains unchanged after a targeted intervention.

The primary research question is:

When a Transformer model trained on short sequences is evaluated on substantially longer sequences, which empirically distinguishable failure behaviours occur, where within the output do the errors arise, and do targeted training interventions alter these behaviours?

This question is divided into five supporting questions:

1. Which combinations of task and positional encoding reliably learn the training-length task across multiple random seeds?
2. For configurations that learn the training task, at which unseen sequence length does performance first collapse, and how rapidly does it deteriorate thereafter?
3. How are errors distributed across output positions, carry conditions, and dependency distances, and do these distributions differ between addition, delayed copy, and reverse?
4. Do learned, sinusoidal, rotary, ALiBi, and no explicit positional encodings exhibit different training-fit, seed-stability, or extrapolation behaviours under otherwise matched conditions?
5. Do mixed-length training, curriculum learning, and randomised padding improve unseen-length performance relative to matched original baselines while preserving training-length performance?

The corresponding aim is to construct an evidence-based taxonomy that separates training-length underfitting, seed instability, immediate extrapolation collapse, position-dependent error structure, and no observed improvement under tested mitigation. The purpose of this taxonomy is to provide a more precise vocabulary for reporting length-generalisation experiments than a single statement that a model “fails” at longer sequences.

1.3 Scope and Experimental Boundaries

This dissertation investigates length generalisation using compact Transformer models under controlled train-short/test-long conditions. The main synthetic benchmark is restricted to addition, delayed copy, and reverse, which provide deterministic tasks with controllable sequence lengths. Associative recall was explored during development but excluded from the main comparison because the model did not learn the training task reliably enough to support a valid extrapolation analysis.

The study compares learned, sinusoidal, RoPE, ALiBi, and no explicit positional encodings across three random seeds. Multi-seed experiments are used to establish training fit, seed stability, and baseline eligibility. More detailed position-wise and task-specific analyses use representative seed-42 checkpoints and are explicitly distinguished from the multi-seed results.

The mitigation study is limited to mixed-length training, curriculum learning, and randomised padding. Each intervention is compared with a matched original baseline and is evaluated according to whether it preserves training-length performance while improving performance at unseen lengths. The conclusions therefore concern the tested configurations rather than the entire families of these training methods.

A limited real-world extension uses the SelfRegulationSCP2 time-series classification dataset. Its inclusion is motivated by the broader use of specialised Transformer architectures for long and irregularly sampled time-series data [13]. However, this extension is exploratory and is not presented as a state-of-the-art time-series benchmark or as direct validation of the synthetic failure taxonomy.

The dissertation does not study parameter scaling, pre-trained language models, retrieval systems, recurrent-memory architectures, or specialised context-window extension methods. Attention and error diagnostics are used to characterise observable behaviour, but they are not treated as complete causal explanations of the models' internal computations.

1.4 Contributions and Achievements

The contribution of this dissertation is primarily empirical and methodological. It does not propose a new Transformer architecture. Instead, it develops a controlled framework for distinguishing different forms of poor long-length performance and applies that framework consistently across tasks, positional encodings, random seeds, and training interventions.

The main contribution is therefore an evidence-based failure taxonomy supported by controlled experiments, multi-seed robustness analysis, position-wise diagnostics, attention diagnostics, and matched mitigation comparisons.

The principal contributions and achievements are:

- A reproducible synthetic benchmark covering addition, delayed copy, and reverse with five positional encoding settings and three random seeds, producing 45 main synthetic training runs, alongside 15 real-world extension runs and separate mitigation experiments.
- A baseline-eligibility procedure that separates configurations which reliably learn the training task from training underfitting and seed-dependent success.
- An evaluation framework that combines exact-match accuracy with token-error percentages, degradation curves, and first-unseen-length comparisons.
- Position-wise and task-specific diagnostics that examine where errors occur, including output-position profiles, carry-conditioned addition errors, and dependency-distance errors for reverse.
- A matched comparison of mixed-length training, curriculum learning, and randomised padding against their corresponding original baselines, with each intervention interpreted according to its intended target.
- An evidence-based failure taxonomy that distinguishes training-length underfitting, seed instability, immediate extrapolation collapse, position-dependent error structure, and no observed improvement under tested mitigation.
- A limited real-world extension using SelfRegulationSCP2 to contrast synthetic sequence-generation failures with behaviour on a time-series classification task.

- A reproducible experimental implementation with separate configurations, checkpoints, raw CSV results, derived summaries, diagnostic outputs, and final report figures for traceability.

The main empirical outcome is that every task-encoding configuration meeting the multi-seed baseline-eligibility criterion collapsed at the first unseen length, although the token-error structure differed across tasks. The tested mitigation variants either underfit or produced no observed improvement at unseen lengths when training-length performance was preserved. These findings motivate the dissertation’s emphasis on describing the form and location of failure rather than reporting only that accuracy decreases.

1.5 Dissertation Structure

The remainder of this dissertation is organised as follows. Chapter 2 introduces the technical background and reviews existing research on Transformers, positional encoding, length generalisation, algorithmic evaluation, and long-context sequence modelling. Chapter 3 describes the experimental methodology, including the synthetic tasks, model architecture, baseline-eligibility procedure, evaluation metrics, diagnostic analyses, mitigation experiments, and real-world extension.

Chapter 4 presents the multi-seed synthetic benchmark results and establishes which task-encoding configurations provide valid extrapolation baselines. Chapter 5 analyses the observed failures in greater detail using positional, task-specific, and attention-based diagnostics before comparing the tested mitigation strategies and constructing the failure taxonomy. Chapter 6 reports the limited SelfRegulationSCP2 time-series extension.

Chapter 7 discusses the findings in relation to the research questions and existing literature, together with the study’s limitations and implications. Finally, Chapter 8 summarises the principal conclusions and identifies directions for future work.

Chapter 2

Background

2.1 Sequence Modelling and Transformer Architecture

Sequence modelling concerns data whose interpretation depends on the ordering and interaction of its elements. Given an input sequence $\mathbf{x} = (x_1, \dots, x_n)$, a sequence model constructs representations that can be used to predict an output at each position, classify the complete sequence, or generate another sequence. Earlier neural sequence-to-sequence systems commonly used recurrent architectures, in which a hidden state was updated sequentially as each input element was processed [20]. This recurrence provided an explicit notion of order, but it also created a sequential computational path through the input.

Attention mechanisms introduced a different way of representing dependencies. Instead of relying exclusively on a single recurrent state, an attention mechanism computes a weighted combination of representations according to their relevance to the current prediction [4]. The Transformer extended this principle by replacing recurrence with stacked self-attention and feed-forward layers [22]. As a result, representations at different sequence positions can be updated in parallel, while each position can directly incorporate information from other positions.

2.1.1 Sequence Representation and Self-Attention

Let the input to a self-attention layer be represented by $\mathbf{X} \in \mathbb{R}^{n \times d_{\text{model}}}$, where n is the sequence length and d_{model} is the representation dimension. Self-attention constructs query, key, and value matrices through learned linear projections:

$$\mathbf{Q} = \mathbf{X}\mathbf{W}^Q, \quad \mathbf{K} = \mathbf{X}\mathbf{W}^K, \quad \mathbf{V} = \mathbf{X}\mathbf{W}^V. \quad (2.1)$$

Here, $\mathbf{W}^Q$, $\mathbf{W}^K$, and $\mathbf{W}^V$ are learned projection matrices, while $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ denote the resulting query, key, and value matrices.

The query associated with one position is compared with the keys associated with all positions. Scaled dot-product attention is then calculated as

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} \right) \mathbf{V}, \quad (2.2)$$

where d_k is the dimensionality of the key vectors [22]. The scaling factor $\sqrt{d_k}$ limits the magnitude of the dot products before the softmax operation. Each row of the resulting attention matrix defines a distribution over the positions from which a query receives information.

This mechanism provides a short computational path between any pair of positions. However, self-attention without additional positional information is insensitive to the order in which the input representations are supplied: permuting the input positions produces the corresponding permutation of the outputs. Consequently, the Transformer requires either explicit positional representations or an architectural mechanism through which position can affect the attention calculation. This requirement is central to the length-generalisation problem considered in this dissertation.

2.1.2 Multi-Head Attention and Transformer Encoder Blocks

Multi-head attention applies the attention calculation using multiple sets of learned projections. For attention head h ,

$$\text{head}_h = \text{Attention}(\mathbf{X}\mathbf{W}_h^Q, \mathbf{X}\mathbf{W}_h^K, \mathbf{X}\mathbf{W}_h^V). \quad (2.3)$$

The head outputs are concatenated and projected back to the model dimension:

$$\text{MultiHead}(\mathbf{X}) = \text{Concat}(\text{head}_1, \dots, \text{head}_{N_{\text{head}}}) \mathbf{W}^O, \quad (2.4)$$

where N_{head} is the number of attention heads and $\mathbf{W}^O$ is a learned output projection. Using separate projections allows different heads to form different attention distributions, although the existence of multiple heads does not guarantee that they learn distinct or interpretable functions.

A Transformer encoder block combines multi-head self-attention with a position-wise feed-forward network. Residual connections preserve the input to each sublayer, while layer normalisation is applied to stabilise the representations. In the post-normalisation formulation used by the original Transformer, the two principal operations can be written as

$$\mathbf{H} = \text{LayerNorm}(\mathbf{X} + \text{MultiHead}(\mathbf{X})), \quad (2.5)$$

$$\mathbf{Z} = \text{LayerNorm}(\mathbf{H} + \text{FFN}(\mathbf{H})), \quad (2.6)$$

with

$$\text{FFN}(\mathbf{H}) = \sigma(\mathbf{H}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2, \quad (2.7)$$

Here, $\mathbf{H}$ is the representation produced by the attention sublayer and $\mathbf{Z}$ is the output of the encoder block. LayerNorm denotes layer normalisation, while FFN denotes the position-wise feed-forward network. The matrices $\mathbf{W}_1$ and $\mathbf{W}_2$ and the vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ are learned parameters, and σ is a non-linear activation function. The same feed-forward transformation is applied independently at every sequence position.

The synthetic models used in this dissertation employ compact stacks of encoder blocks following this general structure. Their precise dimensions, activation function, optimisation settings, and output mappings are reported in Chapter 3. The next section examines the mechanisms through which positional information is incorporated into these blocks.

2.2 Positional Information in Transformers

Unlike recurrent models, self-attention does not process tokens through an inherently ordered sequence of state updates. Without an additional source of positional information, the same token representations presented in a different order produce correspondingly permuted outputs. A Transformer must therefore obtain information about order either through explicit position vectors, modifications to the attention calculation, or structural features such as an autoregressive mask.

Positional methods can broadly be distinguished by whether they represent absolute positions or relationships between positions. Absolute methods associate a representation with each position in the sequence. Relative methods instead modify attention according to the distance or displacement between a query and a key [18]. This distinction is particularly relevant to length generalisation because a model may be evaluated at absolute positions or relative distances that were not observed during training.

2.2.1 Learned and Sinusoidal Positional Encodings

Learned absolute positional embeddings associate each position p with a trainable vector $\mathbf{p}_p$. The input representation at that position is

$$\mathbf{z}_p = \mathbf{e}(x_p) + \mathbf{p}_p, \quad (2.8)$$

where $\mathbf{e}(x_p)$ is the embedding of the token at position p . Because the positional vectors are optimised jointly with the model, they can adapt to the positions represented in the training data. However, this approach does not impose a functional relationship between neighbouring positions. Entries allocated for positions beyond the training range receive no direct training signal, and positions outside the allocated embedding table cannot be represented without extending the table.

The original Transformer also introduced fixed sinusoidal positional encodings [22]. For position p and representation dimension i , these are defined as

$$\text{PE}(p, 2i) = \sin\left(\frac{p}{10000^{2i/d_{\text{model}}}}\right), \quad (2.9)$$

$$\text{PE}(p, 2i + 1) = \cos\left(\frac{p}{10000^{2i/d_{\text{model}}}}\right). \quad (2.10)$$

Here, PE denotes the positional encoding, p is the sequence position, and i indexes pairs of representation dimensions.

Different dimensions therefore vary at different frequencies. Unlike a learned lookup table, the sinusoidal representation is generated by a fixed function and can be evaluated at positions beyond those used during training. Relative offsets can also be expressed through linear transformations of these sinusoidal components. These properties make sinusoidal encoding a plausible candidate for length extrapolation, but they do not guarantee that the surrounding Transformer learns a computation that uses the representation systematically at unseen lengths.

2.2.2 Rotary Position Embedding

Rotary Position Embedding (RoPE) incorporates position into the query and key vectors rather than adding a positional vector to the token representation [19]. Pairs of query and key dimensions are rotated by an angle determined by their position. For position p , the transformation can be written abstractly as

$$\tilde{\mathbf{q}}_p = \mathbf{R}_p \mathbf{q}_p, \quad \tilde{\mathbf{k}}_p = \mathbf{R}_p \mathbf{k}_p, \quad (2.11)$$

where $\mathbf{R}_p$ is a block-diagonal rotation matrix containing position-dependent sine and cosine terms. Attention is subsequently computed using the rotated query and key vectors:

$$\tilde{\mathbf{q}}_p^\top \tilde{\mathbf{k}}_r = \mathbf{q}_p^\top \mathbf{R}_p^\top \mathbf{R}_r \mathbf{k}_r. \quad (2.12)$$

Because the product $\mathbf{R}_p^\top \mathbf{R}_r$ depends on the relative displacement between p and r , RoPE introduces relative-position information into the attention score while retaining an absolute position-dependent transformation. The rotations can be evaluated at longer positions, although their mathematical availability does not ensure that a model trained over a shorter range will interpret the unseen rotation phases correctly.

2.2.3 Attention with Linear Biases

Attention with Linear Biases (ALiBi) represents distance by adding a head-specific linear penalty directly to each attention score [15]. A general form of the biased score between positions i and j in head h is

$$a_{ij}^{(h)} = \frac{\mathbf{q}_i^{(h)} \left(\mathbf{k}_j^{(h)} \right)^\top}{\sqrt{d_k}} - m_h \Delta(i, j), \quad (2.13)$$

where m_h is a fixed slope and $\Delta(i, j)$ is the positional distance used by the attention mechanism. Different slopes cause different heads to apply different distance penalties. In a bidirectional encoder, $\Delta(i, j)$ can be defined using absolute distance, whereas a causal model restricts attention to preceding positions.

ALiBi does not require a positional embedding table and its bias can be calculated at arbitrary sequence lengths. It was specifically introduced under a train-short/test-long evaluation setting. Nevertheless, the linear bias favours particular attention-distance patterns; whether those patterns support extrapolation depends on the task and on the dependencies required to produce the correct output.

2.2.4 Transformers Without Explicit Positional Encoding

A Transformer without explicit positional encoding is commonly described as NoPE. Under this condition, no positional vector is added to the token embedding, and neither RoPE nor a positional attention bias is applied. Research on causal Transformers has shown that models can sometimes recover useful position-related behaviour without an explicit positional encoding, partly because the causal attention mask itself introduces an ordering structure [9].

This observation does not mean that explicit position information is unnecessary in every architecture. In an unmasked bidirectional self-attention encoder, removing all positional mechanisms leaves the model permutation-equivariant: reordering the input tokens causes the outputs to be reordered in the same way, without otherwise identifying their original positions. NoPE is consequently an important control condition. It tests whether task content and the remaining architecture provide enough information to learn the required transformation.

The five settings considered in this dissertation therefore introduce position at different points in the computation. Learned and sinusoidal methods modify the input representations; RoPE transforms queries and keys; ALiBi modifies the attention scores; and NoPE supplies no explicit position signal. Their structural differences motivate the controlled comparison reported in the later experimental chapters.

2.3 Length Generalisation

2.3.1 Length Generalisation and Out-of-Distribution Evaluation

Consider a family of tasks in which f_n denotes the required mapping for sequences of length n . A model is trained using examples whose lengths belong to a training set $\mathcal{L}_{\text{train}}$, with maximum observed length

$$L_{\text{max}} = \max \mathcal{L}_{\text{train}}. \quad (2.14)$$

An ordinary in-distribution evaluation measures performance on new examples whose lengths remain within $\mathcal{L}_{\text{train}}$. In contrast, a length-generalisation evaluation uses test lengths

$$n > L_{\text{max}}, \quad (2.15)$$

while preserving the underlying task definition. The inputs are therefore longer, but the rule required to produce the output remains unchanged.

This setting differs from standard independent and identically distributed testing. The model must extrapolate along a controlled dimension of the data rather than generalise only to new samples drawn from the same length distribution. Strong performance at L_{max} demonstrates that the model can fit the observed task instances, but it does not establish that the learned computation represents the complete family $\{f_n\}$. A solution can be correct over the observed range while depending on length-specific features, absolute positions, or statistical regularities that cease to hold when $n > L_{\text{max}}$.

Length generalisation is related to systematic generalisation: both concern whether learned components or operations can be applied in combinations outside the training distribution. Work on compositional sequence learning has shown that high performance on familiar combinations can coexist with poor performance under systematic changes to the test distribution [10]. Length extrapolation provides a particularly controlled instance of this broader problem because the task semantics can be held constant while only the size of the instance changes.

A rigorous train-short/test-long experiment must therefore separate two questions. First, did the model reliably learn the task at the training length? Second, conditional on that success, did the learned solution transfer to longer instances? Without the first condition, poor long-length accuracy cannot be attributed specifically to extrapolation.

2.3.2 Length Generalisation in Algorithmic Tasks

Algorithmic tasks provide a controlled environment for studying this form of generalisation. Their target outputs are produced by deterministic procedures, and new examples can be generated at precisely specified lengths. In principle, an algorithm such as digit-wise addition or sequence reversal can be applied to inputs longer than those used during training. In practice, however, a neural model can minimise its training loss using a representation that is only reliable over the observed range.

This distinction has motivated research into whether successful neural models learn transferable procedures or length-specific approximations. Studies of arithmetic with Transformer-based language models report that increasing model scale alone does not reliably produce length extrapolation, while intermediate reasoning steps can improve some settings [1]. More broadly, experiments organised according to formal-language complexity show that fitting the training distribution does not imply non-trivial generalisation to longer strings, and that performance depends on both the task structure and the inductive bias of the architecture [7].

Zhou et al. examine this question through the algorithms representable by Transformers and distinguish tasks for which training tends to recover a simple length-generalising program from tasks for which models favour non-generalising pattern-matching solutions [23]. Their results are important because they show that architectural expressivity is not sufficient by itself: the existence of a Transformer implementation of an algorithm does not guarantee that optimisation will recover that implementation from finite training examples.

Synthetic algorithmic benchmarks are therefore useful not because they represent the full complexity of practical applications, but because they make the required rule and the distribution shift explicit. They permit controlled comparisons across tasks, encodings, training procedures, and random seeds. Their simplicity also makes it possible to analyse individual output positions and task-specific dependencies after aggregate accuracy has deteriorated.

2.3.3 Length Generalisation Versus Long-Context Modelling

Length generalisation should not be treated as synonymous with long-context modelling. A long-context architecture is designed to process or retain information across a large input. Transformer-XL, for example, extends the effective context available to a model through segment-level recurrence and a relative positional formulation [6]. The Long Range Arena benchmark instead evaluates the efficiency and effectiveness of architectures on several tasks containing long sequences [21].

These approaches address important computational and representational questions, but operating on a long sequence does not necessarily demonstrate extrapolation. A model can be trained and tested on similarly long examples, in which case the evaluation remains within the observed length regime. Conversely, a compact model can be used in a strict length-generalisation experiment even when its test sequences are modest by contemporary long-context standards, provided that they extend substantially beyond the training range.

The two research areas can nevertheless interact. Positional mechanisms, attention-distance biases, memory structures, and computational cost all influence how a model behaves as its input grows. The defining feature of a length-generalisation experiment is therefore not the absolute magnitude of the context window, but the controlled separation between the lengths used for training and those used for evaluation.

2.4 Controlled Sequence Tasks

Controlled sequence tasks make it possible to vary sequence length without changing the rule that determines the correct output. Unlike natural-language datasets, their target mappings are deterministic and their relevant dependencies can be stated explicitly. This makes them useful for separating failure to learn a task from failure to apply a learned computation at longer lengths.

The tasks used in this dissertation represent three different computational requirements. Addition requires symbolic computation with possible carry propagation; delayed copy requires retention and aligned retrieval after an intervening delay; and reverse requires a position-dependent reordering of the complete input. The following subsections describe the computational properties of these tasks. Their precise data-generation and serialisation procedures are specified in Chapter ??.

2.4.1 Addition

Addition is frequently used as a controlled test of whether neural models can learn arithmetic procedures rather than associations limited to the numerical range represented during training [1, 11]. Consider two n -digit decimal operands with digit sequences

$$\mathbf{a} = (a_1, \dots, a_n), \quad \mathbf{b} = (b_1, \dots, b_n), \quad (2.16)$$

where a_1 and b_1 are the most significant digits. The corresponding integers are

$$A = \sum_{i=1}^n a_i 10^{n-i}, \quad B = \sum_{i=1}^n b_i 10^{n-i}. \quad (2.17)$$

Digit-wise addition proceeds from the least significant position towards the most significant position. If c_{i+1} denotes the carry entering position i , then

$$s_i = a_i + b_i + c_{i+1}, \quad (2.18)$$

$$y_i = s_i \bmod 10, \quad (2.19)$$

$$c_i = \left\lfloor \frac{s_i}{10} \right\rfloor, \quad (2.20)$$

with $c_{n+1} = 0$. Here, y_i is the output digit at position i , while c_i is the carry propagated towards the next more significant position.

The task therefore combines aligned local interactions with a dependency that can propagate across several digit positions. A model may perform well on short operands by learning statistical or position-specific regularities without recovering a procedure that remains reliable for longer operands. Furthermore, aggregate accuracy alone does not reveal whether failures are associated with carry propagation, output alignment, or a broader collapse across all positions. These alternatives motivate the carry-conditioned and position-wise diagnostics used later in the dissertation.

2.4.2 Delayed Copy

Copying tasks have been used to evaluate whether sequence models can preserve information across an intervening delay [2]. Given a symbol sequence

$$\mathbf{x} = (x_1, \dots, x_n), \quad (2.21)$$

the delayed-copy task first presents the sequence, followed by a separator and a region of padding symbols. The required output reproduces the original symbols after this delay. Abstractly, the input and target can be represented as

$$\mathbf{x}_{\text{input}} = (x_1, \dots, x_n, \text{SEP}, \underbrace{\text{PAD}, \dots, \text{PAD}}_n), \quad (2.22)$$

$$\mathbf{x}_{\text{target}} = (\underbrace{\text{PAD}, \dots, \text{PAD}}_{n+1}, x_1, \dots, x_n). \quad (2.23)$$

Here, **SEP** marks the boundary between the source and delayed-output regions, and **PAD** is a padding symbol. Each predicted content token must match a corresponding token from the first part of the input.

Increasing n simultaneously increases the amount of information that must be retained and the distance between each source symbol and its required output position. The task does not require arithmetic transformation, but it does require reliable content preservation and positional alignment. It therefore provides a controlled contrast with addition: errors can indicate a failure to retrieve or align stored symbols rather than a failure of symbolic calculation.

2.4.3 Reverse

For the reverse task, an input sequence

$$\mathbf{x} = (x_1, x_2, \dots, x_n) \quad (2.24)$$

is mapped to

$$\mathbf{y} = (x_n, x_{n-1}, \dots, x_1), \quad y_i = x_{n-i+1}. \quad (2.25)$$

Unlike delayed copy, reverse does not preserve the original positional alignment. Every output position must identify the input position determined by the sequence length. The required source position for a fixed output index therefore changes when n changes. This makes reverse a direct test of whether a model has learned a general position-dependent transformation rather than associations tied to absolute training positions.

The source-target dependency distance also varies across the output. Tokens near the ends of the sequence must be transferred across large distances, whereas positions near the centre have shorter dependencies. This structure allows performance to be analysed according to dependency distance rather than only output position. Prior controlled studies have similarly used formal and algorithmic sequence

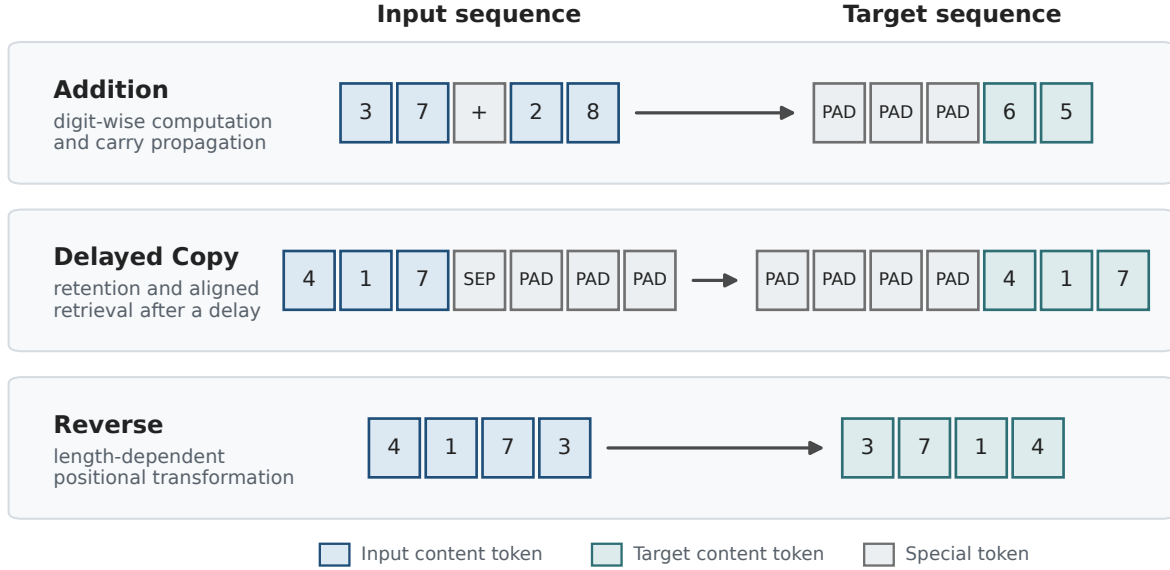


Figure 2.1: Representative input–target mappings for addition, delayed copy, and reverse. Addition combines digit-wise computation with carry propagation; delayed copy requires retrieval after an intervening padded region; and reverse maps each output position to its length-dependent source position. PAD and SEP denote padding and separator tokens, respectively. The examples illustrate task serialisation rather than the sequence lengths used in the experiments.

transformations to examine whether neural architectures transfer learned computations to longer inputs [7, 23].

Figure 2.1 summarises representative input–target mappings for the three controlled tasks.

Together, the three tasks separate computational properties that would be conflated in a single aggregate benchmark. Addition emphasises computation and carry propagation, delayed copy emphasises retention and aligned retrieval, and reverse emphasises length-dependent positional transformation. This separation supports a failure analysis that asks not only whether accuracy decreases, but whether different task structures produce different error behaviours.

2.5 Evaluating Length-Generalisation Failure

A length-generalisation evaluation must establish more than whether aggregate accuracy decreases. It must first verify that the model learned the training-length task, then identify where deterioration begins and how errors are distributed. Sequence-level accuracy, token-level accuracy, random-seed variation, position-wise profiles, and attention statistics provide complementary evidence about these behaviours.

2.5.1 Sequence-Level and Token-Level Metrics

Let N_L denote the number of evaluation examples at test length L . After applying any task-specific padding mask, let $\mathbf{y}^{(r)}$ and $\hat{\mathbf{y}}^{(r)}$ denote the evaluated target and predicted token sequences for example r . Exact-match accuracy is

$$A_{\text{EM}}(L) = \frac{100}{N_L} \sum_{r=1}^{N_L} \mathbb{I} \left[\hat{\mathbf{y}}^{(r)} = \mathbf{y}^{(r)} \right], \quad (2.26)$$

where $\mathbb{I}[\cdot]$ is the indicator function. An example contributes to the exact-match score only when every predicted token is correct. Exact match therefore measures complete task success, but it does not show how many tokens were incorrect within a failed sequence.

If target example r contains T_r evaluated tokens, token accuracy is

$$A_{\text{tok}}(L) = \frac{100}{\sum_{r=1}^{N_L} T_r} \sum_{r=1}^{N_L} \sum_{j=1}^{T_r} \mathbb{I} \left[\hat{y}_j^{(r)} = y_j^{(r)} \right]. \quad (2.27)$$

Token accuracy distinguishes almost-correct predictions from predictions that are incorrect across nearly every output position. The corresponding error percentages are

$$E_{\text{EM}}(L) = 100 - A_{\text{EM}}(L), \quad (2.28)$$

$$E_{\text{tok}}(L) = 100 - A_{\text{tok}}(L). \quad (2.29)$$

Reporting errors as percentages makes the magnitude of failure directly interpretable. For example, zero exact-match accuracy always corresponds to 100% exact-match error, but the token-error percentage may range from a small value to a value close to uniform prediction.

Exact match also becomes increasingly strict as the output grows. If token predictions were independent and each token were correct with probability p , the probability of an entirely correct sequence of length T would be

$$P(\text{exact match}) = p^T. \quad (2.30)$$

This expression is an illustration rather than an independence assumption about the models. It shows why sequence-level performance can fall sharply even when token accuracy decreases more gradually.

For a target vocabulary $\mathcal{V}_{\text{target}}$, uniform prediction gives expected token accuracy $100/|\mathcal{V}_{\text{target}}|$. The corresponding token-error reference is

$$E_{\text{chance}} = 100 \left(1 - \frac{1}{|\mathcal{V}_{\text{target}}|} \right). \quad (2.31)$$

This value provides a descriptive reference for near-uniform prediction; it is not, by itself, a statistical significance test.

2.5.2 Random-Seed Stability

Neural-network training is stochastic because parameter initialisation, data ordering, and optimisation can vary between runs. Reporting only the best-performing seed can therefore conceal unstable training behaviour. Empirical studies have demonstrated that random seeds can materially change the conclusions drawn from sequence-model experiments, motivating the reporting of score distributions or multiple-run summaries [16].

For a metric $A_s(L)$ obtained from seed s , the mean across S seeds is

$$\bar{A}(L) = \frac{1}{S} \sum_{s=1}^S A_s(L), \quad (2.32)$$

and the sample standard deviation is

$$\text{SD}(L) = \sqrt{\frac{1}{S-1} \sum_{s=1}^S (A_s(L) - \bar{A}(L))^2}. \quad (2.33)$$

In this dissertation, error bars representing $\bar{A}(L) \pm 1 \text{SD}(L)$ describe variation across random seeds. They do not represent standard errors or confidence intervals. Because the main benchmark uses three seeds, individual seed outcomes are also retained when assessing whether a configuration learned the training task reliably.

This distinction supports the baseline-eligibility procedure introduced in Chapter ???. A configuration that succeeds for only one initialisation is evidence of seed instability, whereas a configuration that succeeds consistently at the training length but fails at longer lengths provides a valid basis for studying extrapolation.

2.5.3 Position-Wise and Task-Specific Diagnostics

Aggregate token error can conceal substantial variation across an output. Let N_j be the number of evaluated examples containing output position j . The position-wise token-error percentage is

$$E_j = 100 \left[1 - \frac{1}{N_j} \sum_{r=1}^{N_j} \mathbb{I} \left[\hat{y}_j^{(r)} = y_j^{(r)} \right] \right]. \quad (2.34)$$

Plotting E_j across positions reveals whether errors are approximately uniform, concentrated near sequence boundaries, or progressively increasing through the output. Positions can also be grouped into start, middle, and end regions to provide compact percentage summaries without discarding the full position-wise profile.

Task-specific groupings test more focused hypotheses. For addition, output digits can be divided according to whether they receive an incoming carry. For reverse, errors can be grouped by the distance between each output position and its required source position. Delayed copy can be examined according to the aligned source and target locations. These analyses identify associations between error and task structure, although they do not alone establish the internal cause of a prediction.

2.5.4 Attention-Based Diagnostics

Attention weights provide an additional description of how the model distributes attention mass. Let $\alpha_{ij}^{(h)}$ denote the attention weight from query position i to key position j in head h , with

$$\sum_{j=1}^n \alpha_{ij}^{(h)} = 1. \quad (2.35)$$

The normalised attention entropy is

$$H_i^{(h)} = - \frac{\sum_{j=1}^n \alpha_{ij}^{(h)} \log \alpha_{ij}^{(h)}}{\log n}. \quad (2.36)$$

Values closer to zero indicate concentrated attention, while values closer to one indicate a more diffuse distribution. Dividing by $\log n$ makes entropy more comparable across sequence lengths.

The average attention distance is

$$D_i^{(h)} = \sum_{j=1}^n \alpha_{ij}^{(h)} |i - j|, \quad (2.37)$$

and the local-attention ratio within a window of radius w is

$$R_i^{(h)}(w) = \sum_{\substack{j=1 \\ |i-j| \leq w}}^n \alpha_{ij}^{(h)}. \quad (2.38)$$

These quantities describe whether attention becomes more diffuse, operates over greater distances, or remains concentrated near each query position as sequence length increases. They must nevertheless be interpreted alongside accuracy. Average distance can grow mechanically when longer positions become available, and a change in attention statistics does not necessarily identify the cause of a model's error.

Attention weights are therefore treated in this dissertation as behavioural diagnostics rather than complete causal explanations. Prior research has shown that substantially different attention distributions can sometimes produce similar predictions, cautioning against interpreting attention weights as faithful explanations of model decisions [8]. Their role here is to identify changes associated with extrapolation failure, not to claim that an observed attention pattern alone caused that failure.

2.6 Training Strategies for Length Generalisation

Training interventions for length generalisation address different possible sources of failure. Exposing a model to several training lengths targets dependence on a single length distribution. Curriculum learning targets the optimisation path by increasing task difficulty gradually. Randomised positional exposure

instead targets the possibility that the positional representations encountered at longer lengths are out of distribution.

These strategies should not be interpreted as interchangeable. An intervention may improve optimisation without changing extrapolation, or it may expose additional positions while weakening performance on the original training task. Its intended target and its training-length performance must therefore be established before unseen-length results are interpreted.

2.6.1 Mixed-Length Training

A fixed-length training condition draws every training example from a single sequence length L_{train} . Mixed-length training instead draws examples from a set

$$\mathcal{L}_{\text{mix}} = \{L_1, L_2, \dots, L_K\}, \quad (2.39)$$

where the lengths are no greater than the maximum training length. If π_k is the proportion of training examples drawn from length L_k , the mixed-length objective can be written as

$$\mathcal{J}_{\text{mix}}(\boldsymbol{\theta}) = \sum_{k=1}^K \pi_k \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim \mathcal{D}_{L_k}} [\ell(f_{\boldsymbol{\theta}}(\mathbf{x}), \mathbf{y})], \quad (2.40)$$

where $\boldsymbol{\theta}$ denotes the model parameters, $\mathcal{D}_{L_k}$ is the data distribution at length L_k , and ℓ is the training loss.

The hypothesis is that exposure to multiple lengths discourages a model from associating the task with one fixed sequence shape or one set of boundary positions. It may instead encourage features that are shared across the observed length range. This motivation is consistent with evidence that Transformer length generalisation depends on both the task and the training distribution [1, 23].

Mixed-length training introduces a trade-off. For a fixed total data budget, including more lengths reduces the number of examples available at each individual length. Padding and batching can also change the proportion of content and special tokens seen during optimisation. Consequently, a mixed-length model may underfit even the original training task. Any apparent extrapolation benefit must therefore be considered together with training-length accuracy and a matched fixed-length baseline.

2.6.2 Curriculum Learning

Curriculum learning presents training examples in a progression from simpler to more difficult conditions [5]. In a length-based curriculum, sequence length acts as the difficulty variable. A model can be trained through an ordered series

$$L_1 < L_2 < \dots < L_K, \quad (2.41)$$

where the parameters learned at stage $k - 1$ initialise training at stage k . Abstractly,

$$\boldsymbol{\theta}_k = \text{Train}(\boldsymbol{\theta}_{k-1}, \mathcal{D}_{L_k}). \quad (2.42)$$

The intended target is optimisation. Shorter examples may allow the model to learn a partial or simpler form of the required computation before longer dependencies are introduced. The later stages then refine the existing parameters rather than learning the longest training task entirely from a random initialisation.

A successful curriculum does not necessarily produce extrapolation. The model may retain useful features from earlier stages, but it may also adapt to the final curriculum length and remain dependent on that observed range. Curriculum learning must therefore be compared with a model trained directly at the final training length. The comparison tests whether gradual length increases alter the resulting behaviour, rather than assuming that the curriculum is intrinsically beneficial.

2.6.3 Randomised Positional Exposure and Padded Training

A separate hypothesis is that failure occurs because positional representations at unseen lengths differ from those encountered during training. Ruoss et al. propose randomised positional encodings that simulate a larger positional range during training [17]. For a sequence of length n , their approach selects an ordered subset

$$p_1 < p_2 < \dots < p_n \tag{2.43}$$

This formulation describes the randomised positional-index method proposed by Ruoss et al.; it is not the padded mixed-length implementation evaluated in this dissertation.

from a larger range of possible indices and applies the positional representation associated with each selected p_i . This allows short training sequences to encounter positional indices that would ordinarily appear only in longer inputs. Their method directly targets positional out-of-distribution exposure.

Padding-based strategies are related but not equivalent. Presenting examples of several logical lengths within a common padded tensor can vary the locations of separators, output regions, and sequence boundaries. However, right-padding alone does not randomly change the starting position of the content, nor does it reproduce randomised positional-index sampling. Padding-based training should consequently be described according to the positions it actually varies.

The padded condition evaluated in this dissertation is motivated by the broader positional-exposure hypothesis, but is treated as a distinct intervention rather than a reproduction of randomised positional encoding. Its precise construction, including the use of mixed logical lengths, right-padding, and shuffled training examples, is described in Chapter

2.6.4 Intervention-Specific Evaluation

The effect of an intervention at length L can be expressed relative to its matched original baseline as

$$\Delta A(L) = A_{\text{intervention}}(L) - A_{\text{baseline}}(L). \tag{2.44}$$

A positive value at unseen lengths is meaningful only if the intervention also preserves adequate performance at the training length. If the intervention fails to learn the training task, the result indicates training-length underfitting rather than evidence about improved or worsened extrapolation.

The three strategy families therefore test different hypotheses: mixed-length training tests whether broader length exposure reduces dependence on a single training length; curriculum learning tests whether a gradual optimisation path produces a more transferable solution; and the padded mixed-length condition tests whether varying logical boundaries and output positions within a common tensor changes extrapolation behaviour. This intervention-specific framing prevents all unsuccessful variants from being assigned the same interpretation.

2.7 Long Sequences in Time-Series Classification

Multivariate time-series classification assigns a single class label to a sequence containing several measurements at each time step. This differs fundamentally from the controlled sequence-to-sequence tasks considered in this dissertation. In Addition, Delayed Copy, and Reverse, increasing the input length also increases the number of output tokens that must be predicted correctly. In time-series classification, by contrast, the output remains a single class label regardless of the input length. Consequently, length-related behaviour is evaluated using classification accuracy rather than token or exact-match accuracy.

Transformers are relevant to long time series because self-attention can model interactions between observations separated by many time steps. However, long sequences also increase the computational cost of attention and may contain dependencies at multiple temporal scales. Real-world time series can introduce additional complications, including irregular sampling, noise, and variation in the quantity of information available across a sequence. Specialised approaches such as Rough Transformers therefore use continuous-time representations and signature-based attention to capture local and global temporal structure while reducing computational requirements [13]. Rough Transformers are discussed here as motivation for studying length in real-world time-series models; they are not implemented in the experiments in this dissertation.

The exploratory real-world experiment uses SelfRegulationSCP2 from the UEA Multivariate Time Series Classification Archive [3]. The archive was introduced to support reproducible comparison of multivariate time-series classification methods and provides predefined training and test splits across datasets from different application domains. SelfRegulationSCP2 provides a useful contrast to the deterministic synthetic tasks because each example contains multiple real-valued channels and is associated with a sequence-level class label.

In this dissertation, a compact Transformer is trained using a fixed prefix of each training series and subsequently evaluated using progressively longer prefixes. The same five positional-encoding conditions

used in the controlled experiments are considered. This design examines whether the classifier’s performance is sensitive to input lengths beyond the prefix used during training. Nevertheless, it should not be interpreted as an exact equivalent of algorithmic length extrapolation. A longer time-series prefix can introduce additional useful evidence, irrelevant observations, or noise, whereas a longer synthetic example applies the same deterministic rule to more symbols.

The real-world experiment is therefore treated as a limited exploratory extension rather than direct validation of the synthetic-task failure taxonomy. Its purpose is to examine whether length sensitivity also appears in a noisy sequence-level classification setting and whether its empirical form differs from the immediate exact-match collapse observed on the controlled tasks.

2.8 Research Gap and Chapter Summary

2.8.1 Research Gap

Previous work has established that positional representation and training length can substantially affect Transformer extrapolation. ALiBi was designed to improve evaluation on sequences longer than those used during training [15], while comparative studies have shown that the choice of positional encoding can alter length-generalisation behaviour [9]. Other work has investigated algorithmic tasks and randomised positional strategies as ways of improving generalisation beyond the observed training lengths [23, 17].

However, reporting only accuracy at longer lengths does not fully explain a model’s behaviour. A low out-of-distribution score may result from several different situations: the model may not have reliably learned the training-length task, performance may be unstable across random seeds, failure may occur immediately after the training boundary, or errors may be concentrated in particular output positions or task-specific conditions. These cases have different interpretations and should not be grouped together as a single form of length-generalisation failure.

There is therefore a need for an evaluation framework that separates baseline validity from extrapolation behaviour. In particular, the training-length baseline should be verified across multiple seeds before conclusions are drawn from longer test sequences. Once a valid baseline has been established, the analysis should examine not only whether accuracy decreases, but also where errors occur, how close performance is to a task-appropriate chance reference, and whether the same behaviour is reproduced across positional encoding choices.

Mitigation experiments require similar care. An intervention can only be interpreted as improving length generalisation if it first preserves adequate performance at the training length. Otherwise, low out-of-distribution performance may simply reflect training underfit. Furthermore, mixed-length training, curriculum learning, and randomised padding target different possible causes of failure. Their outcomes should consequently be compared with matched original baselines and interpreted in relation to the specific hypothesis each intervention was intended to test.

This dissertation addresses these gaps through a controlled comparison of five positional-encoding settings across Addition, Delayed Copy, and Reverse. It combines multi-seed baseline eligibility with exact-match and token-level metrics, percentage-based error reporting, position-wise diagnostics, task-specific analyses, and matched baseline-versus-mitigation comparisons. A limited multivariate time-series classification experiment is included as an exploratory contrast, while remaining separate from the main synthetic failure taxonomy.

2.8.2 Chapter Summary

This chapter introduced the Transformer components required for the study, including self-attention and positional representation. It explained why performance at the training length is insufficient evidence of length generalisation and described the distinct computational demands of Addition, Delayed Copy, and Reverse. It also established the roles of exact-match accuracy, token error, multi-seed evaluation, position-wise analysis, and task-specific diagnostics.

The chapter further distinguished the hypotheses underlying mixed-length training, curriculum learning, and randomised padding, and explained why a mitigation must retain training-length performance before its extrapolation behaviour can be interpreted. Finally, it situated the exploratory SelfRegulation-SCP2 experiment within multivariate time-series classification and clarified why sequence-level classification is not directly equivalent to deterministic sequence-to-sequence extrapolation.

The next chapter translates these principles into the experimental methodology, specifying the datasets,

model architecture, positional-encoding conditions, training procedure, baseline-eligibility criterion, evaluation metrics, diagnostic analyses, and mitigation comparisons.

Chapter 3

Methodology and Experimental Design

This chapter specifies the experimental methodology used to investigate length generalisation in Transformer-based sequence models. The experiments follow a controlled train-short/test-long protocol in which a model is trained using sequences of a fixed length and subsequently evaluated on progressively longer unseen sequences. This design makes it possible to distinguish successful learning at the training length from extrapolation beyond the observed length distribution.

The methodology was organised around three requirements. First, a configuration had to learn the training-length task reliably before its longer-length behaviour could be interpreted. Second, conclusions had to be supported across multiple random seeds rather than inferred from a single run. Third, evaluation had to describe not only whether performance decreased, but also where and how the resulting errors were distributed.

The chapter first presents the overall experimental pipeline and then describes the controlled tasks, dataset generation, Transformer architecture, positional-encoding conditions, training procedure, baseline-eligibility criterion, evaluation metrics, diagnostic analyses, and mitigation experiments. A separate exploratory experiment on multivariate time-series classification is presented at the end of the chapter. Throughout the controlled experiments, the architecture and training procedure were kept as consistent as possible so that sequence length, task, positional encoding, and training intervention remained the principal experimental variables.

3.1 Experimental Overview

The study used a staged experimental pipeline. Controlled algorithmic tasks were first generated at predefined training and evaluation lengths. A compact Transformer was then trained for each combination of task, positional encoding, and random seed. Five positional-encoding conditions were examined: learned positional embeddings, sinusoidal encoding, no explicit positional encoding, ALiBi, and RoPE. Each baseline condition was repeated using three random seeds, giving 45 baseline training runs across the three tasks.

The train-short/test-long protocol follows the general form used in previous investigations of positional extrapolation and algorithmic length generalisation [15, 9, 23]. However, the present methodology introduces an explicit baseline-eligibility stage before extrapolation results are interpreted. Training-length performance was assessed separately for every task-encoding-seed combination. Only task-encoding configurations that learned the training task sufficiently reliably across seeds were used as the principal baselines for the subsequent failure analysis.

Eligible baselines were evaluated over a sequence of increasing test lengths. The resulting curves were used to determine whether degradation began at the first unseen length or emerged more gradually. Exact-match accuracy was supplemented by token-error percentages, position-wise error profiles, and task-specific diagnostics. These analyses examined whether errors were distributed uniformly across an output or associated with particular positions, carry conditions, or dependency distances. Attention analyses were used as supporting diagnostics rather than as direct evidence that a model had learned a particular algorithm.

Mitigation experiments were conducted only as matched comparisons with an original baseline using

the same task and positional-encoding condition. Mixed-length training, curriculum learning, and randomised padding were treated as separate interventions because they test different hypotheses about the source of length-related failure. An intervention was not interpreted as improving extrapolation unless it first retained adequate performance at the training length.

Finally, a limited experiment using the SelfRegulationSCP2 multivariate time-series dataset examined input-length sensitivity in a sequence-level classification setting. This experiment used the same positional-encoding conditions and multiple random seeds, but it was analysed separately because time-series classification is not directly equivalent to deterministic sequence-to-sequence extrapolation.

Stage	Procedure	Purpose
1	Controlled data generation	Create deterministic training and test sets at predefined sequence lengths.
2	Multi-seed baseline training	Train each task and positional-encoding condition using seeds 42, 123, and 2024.
3	Baseline eligibility	Identify configurations that reliably learn the training-length task before interpreting extrapolation.
4	Extrapolation diagnostics	Measure exact-match and token errors across lengths and determine where failure occurs.
5	Matched mitigation comparison	Compare each intervention with its corresponding original baseline and check whether training-length performance is retained.
6	Exploratory real-world extension	Examine input-length sensitivity on SelfRegulationSCP2 using sequence-level classification accuracy.

Table 3.1: Stages of the experimental methodology.

3.2 Synthetic Task Design

The conceptual motivation for Addition, Delayed Copy, and Reverse was discussed in Chapter 2, and their transformations are illustrated in Figure 2.1. This section therefore focuses on their exact experimental implementation.

For each task, the nominal length L has a task-specific meaning. In Addition, L is the number of digits in each operand; in Delayed Copy, it is the number of content tokens to reproduce; and in Reverse, it is the complete sequence length. The resulting sequence construction is summarised in Table 3.2.

Task	Meaning of L	Encoded sequence length	Vocabulary
Addition	Digits per operand	$2L+1$ input and target tokens	10 digits, plus, and padding
Delayed Copy	Content tokens to reproduce	$2L+1$ input and target tokens	100 content symbols, separator, and padding
Reverse	Number of input tokens	L input and target tokens	100 content symbols

Table 3.2: Implementation of the controlled synthetic tasks.

For Addition, leading zeros were excluded so that both operands always had the intended length. The sum was right-aligned within the target and preceded by padding tokens. For Delayed Copy, the input contained the content sequence, a separator, and L padding tokens; the target contained $L+1$ padding tokens followed by the original content sequence. Reverse directly mapped each generated sequence to the same tokens in the opposite order.

Associative Recall was also implemented during development but was excluded from the main comparison because it did not provide sufficiently reliable training-length performance. This decision followed the baseline-eligibility principle: extrapolation behaviour was not interpreted when the original task had not first been learned reliably.

3.3 Train-Short/Test-Long Protocol

Each model was trained using examples at one nominal length L_{train} and evaluated at both the training length and a sequence of progressively longer lengths [15, 9, 23]. Performance at $L_{\text{test}} = L_{\text{train}}$ was treated as in-distribution performance, while evaluations satisfying $L_{\text{test}} > L_{\text{train}}$ were treated as out-of-distribution length extrapolation.

Addition models were trained using operands containing 16 digits and evaluated using operands containing up to 1024 digits. Delayed Copy and Reverse models were trained at a nominal length of 128 and evaluated at lengths up to 1024. The complete schedule is shown in Table 3.3. These values refer to nominal task length: as specified in Section 3.2, the encoded sequence length is $2L + 1$ for Addition and Delayed Copy and L for Reverse.

Task	L_{train}	First length	unseen	L_{test}
Addition	16	32		16, 32, 64, 128, 256, 512, 1024
Delayed Copy	128	256		128, 256, 512, 1024
Reverse	128	256		128, 256, 512, 1024
SelfRegulationSCP2	256	512		256, 512, 1024, 1152

Table 3.3: Training and evaluation-length schedule. Synthetic-task values denote nominal task lengths, while SelfRegulationSCP2 values denote time-series prefix lengths.

For the synthetic tasks, a separate test set was generated at every evaluation length using the same deterministic rule as the training data. The underlying task therefore remained unchanged; only the length required to execute the task increased.

Training-length performance was evaluated before extrapolation behaviour. A configuration that did not learn the task reliably at L_{train} was classified as training underfit and was not used as evidence of length-generalisation failure. For configurations that satisfied the baseline-eligibility criterion, performance at the first unseen and subsequent lengths was used to characterise whether degradation was immediate or gradual. The numerical eligibility and collapse thresholds are defined in Section 3.8.

3.4 Experiment Matrix

The synthetic baseline experiment crossed three tasks, five positional-encoding conditions, and three random seeds. This produced

$$3 \text{ tasks} \times 5 \text{ positional encodings} \times 3 \text{ seeds} = 45$$

independently trained synthetic baseline models. Each trained checkpoint was evaluated at every test length configured for its task. Associative Recall was not included in this matrix because it did not satisfy the training-fit requirement described in Section 3.2.

The complete synthetic matrix was first used to assess training-length performance and seed stability. Baseline eligibility was then determined at the task-encoding level using results from all three seeds. Consequently, all 45 baseline runs contributed to the eligibility analysis, while the principal extrapolation and failure-behaviour analyses focused on task-encoding configurations that satisfied the eligibility criterion.

The real-world extension crossed one dataset with the same five positional encodings and three seeds, producing a further 15 trained models. It was analysed separately from the synthetic benchmark because it predicts one sequence-level class rather than a target token at every output position.

Table 3.4 summarises the baseline experiment matrix.

The task dimension was used to compare behaviour across distinct deterministic transformations. The positional-encoding dimension examined whether training fit and extrapolation varied with the representation of sequence order. Finally, the seed dimension measured sensitivity to model initialisation, training stochasticity, and synthetic data generation. Mitigation experiments were conducted as additional matched interventions and are specified separately in Section 3.11.

Experiment component	Values	Models
Synthetic tasks	Addition, Delayed Copy, Reverse	–
Synthetic positional encodings	Learned, sinusoidal, NoPE, ALiBi, RoPE	–
Synthetic seeds	42, 123, 2024	–
Synthetic baseline matrix	3 tasks $\times$ 5 encodings $\times$ 3 seeds	45
Real-world dataset	SelfRegulationSCP2	–
Real-world baseline matrix	1 dataset $\times$ 5 encodings $\times$ 3 seeds	15
Total baseline models	Synthetic and real-world experiments	60

Table 3.4: Baseline experimental matrix. The counts refer to independently trained model checkpoints and exclude the subsequent mitigation experiments.

3.5 Model Architecture

The synthetic experiments used a compact encoder-only Transformer implemented from scratch in PyTorch, following the principal components of the original Transformer architecture [22]. The model was not pretrained. It consisted of a task-specific token-embedding layer, one of the five positional treatments, two Transformer blocks, and a linear output layer that produced vocabulary logits independently at every sequence position.

Each Transformer block applied multi-head self-attention followed by a position-wise feedforward network. Residual connections and dropout were applied around both sublayers, with layer normalisation performed after each residual addition. The feedforward network used a ReLU activation between its two linear transformations.

The model used full bidirectional self-attention and did not apply a causal mask. This was appropriate because every controlled task was formulated as a non-autoregressive sequence-to-sequence transformation: the complete input sequence was supplied simultaneously, and the model predicted a target token at every position. The architecture was held constant across positional-encoding conditions, except for the positional component being evaluated.

Architecture component	Configuration
Token-embedding dimension	128
Transformer blocks	2
Attention heads	4
Dimension per attention head	32
Feedforward hidden dimension	256
Feedforward activation	ReLU
Dropout probability	0.1
Normalisation	Post-residual layer normalisation
Attention visibility	Full bidirectional attention; no causal mask
Output layer	Linear projection to task vocabulary at every position

Table 3.5: Transformer architecture used for the controlled synthetic experiments. Optimisation settings are reported separately in Section 3.7.

The maximum supported sequence length was chosen to cover the longest encoded input in each task. It was set to 2049 positions for Addition and Delayed Copy, corresponding to $2(1024) + 1$, and to 1024 positions for Reverse. These limits determined the size of the learned positional-embedding table and the precomputed sinusoidal encoding buffer but did not change the number of Transformer blocks or attention heads.

The architecture was intentionally compact because the objective was controlled comparison rather than state-of-the-art task performance. Keeping the architecture fixed made it possible to repeat the full experiment matrix across tasks, positional encodings, lengths, and seeds while attributing differences more directly to the experimental conditions.

3.6 Positional Encoding Variants

Five positional conditions were evaluated: learned absolute embeddings, sinusoidal encodings, no explicit positional encoding, ALiBi, and RoPE. Their theoretical motivations were discussed in Chapter 2; this section specifies how they were implemented in the experimental model.

Only the positional treatment was changed between conditions. The token embedding, number of Transformer blocks, attention heads, feedforward dimension, output layer, and optimisation procedure otherwise remained fixed. This controlled design reduced the possibility that an architectural change unrelated to position caused the observed performance differences.

Encoding	Application point	Experimental implementation
Learned	Input representations	A trainable embedding vector was added for each absolute position.
Sinusoidal	Input representations	Fixed sine and cosine vectors were added using the original Transformer frequency schedule.
NoPE	No explicit intervention	Token embeddings were passed to the first Transformer block without an added positional signal.
ALiBi	Attention logits	A head-specific negative bias proportional to the absolute distance between query and key positions was added before the softmax.
RoPE	Queries and keys	Pairs of query and key dimensions were rotated using position-dependent sine and cosine frequencies before their dot product was calculated.

Table 3.6: Implementation of the five positional-encoding conditions.

For the learned condition, a trainable positional-embedding table was allocated up to the maximum supported sequence length. During baseline training, however, only entries corresponding to positions present at the training length received gradient updates. Embeddings assigned exclusively to longer unseen positions therefore remained at their initial values. This is an inherent property of the evaluated learned absolute-position baseline and is relevant when interpreting its extrapolation behaviour.

The sinusoidal condition used the fixed sine and cosine construction from the original Transformer [22]. Because these vectors were defined in advance for every supported position, evaluation at a longer length did not require learning additional position parameters.

The NoPE condition applied no explicit positional transformation. It was included because previous work has shown that Transformers without an explicit positional encoding can still exhibit task-dependent positional and length-generalisation behaviour [9]. In the present bidirectional implementation, NoPE otherwise used the same attention and feedforward operations as the other conditions.

ALiBi added a head-specific linear penalty to the attention logits [15]. Because the model used bidirectional rather than causal attention, the implemented bias depended on the absolute separation between positions:

$$B_{ij}^{(h)} = -m_h|i - j|,$$

where m_h is the fixed slope for attention head h . This symmetric form favoured nearby positions in both directions. It should therefore be understood as a bidirectional adaptation of the ALiBi distance bias rather than the original causal-language-model setting.

RoPE applied rotary position information directly to the query and key representations before attention scores were computed [?]. The implementation used an even head dimension of 32 and the standard frequency base of 10 000. Value representations were not rotated.

All five conditions were subjected to the same multi-seed baseline-eligibility assessment. Consequently, a positional condition that failed to learn a task at the training length was classified as training underfit rather than as evidence of extrapolation failure. The eligibility procedure is defined in Section ??.

3.7 Training and Evaluation Procedure

All models were implemented and trained using PyTorch [14]. For each synthetic configuration, the generated data contained 10,000 training examples, 1,000 validation examples, and 1,000 test examples

at every evaluation length. The validation split was generated as part of the data pipeline but was not used for early stopping or checkpoint selection in the final experiments.

Each baseline model was trained for 10 epochs. The checkpoint saved after the final epoch was used for evaluation; no best-validation checkpoint was selected. Optimisation used AdamW [12] with a learning rate of 10^{-3} . No learning-rate scheduler, early stopping, or gradient clipping was applied. Because the optimiser was instantiated using the PyTorch defaults apart from the learning rate, its weight-decay coefficient was 0.01.

The loss was token-level cross-entropy. For Addition and Delayed Copy, target positions containing the padding token were excluded from the loss. Reverse does not use a padding token, so every target position contributed to its training loss.

Training component	Configuration
Training examples per run	10,000
Validation examples	1,000 at the training length
Test examples	1,000 at each evaluation length
Training epochs	10
Optimiser	AdamW
Learning rate	10^{-3}
Weight decay	0.01
Learning-rate schedule	None
Checkpoint selection	Final training epoch
Addition batch size	8
Delayed Copy batch size	1
Reverse batch size	1
Random seeds	42, 123, and 2024

Table 3.7: Training and evaluation configuration for the synthetic baseline experiments.

The three seeds controlled Python, NumPy, and PyTorch randomness, including synthetic data generation and model initialisation. GPU random-number generators were also seeded when CUDA was available. The seed repetitions therefore measured the combined sensitivity of the experimental pipeline to data sampling, parameter initialisation, minibatch ordering, and stochastic training operations.

After training, each checkpoint was loaded in evaluation mode and evaluated separately on the held-out dataset for every configured test length. Dropout was disabled during evaluation, and predictions were obtained using the highest-logit token at each output position. All test sets were generated before evaluation and were not used for model selection or further optimisation.

3.8 Evaluation Metrics

The synthetic tasks were evaluated using token accuracy and exact-match accuracy. Let $y_{n,j}$ denote the target token at position j in sequence n , and let $\hat{y}_{n,j}$ denote the corresponding predicted token. A binary mask $m_{n,j}$ indicates whether that position is included in evaluation. For Addition and Delayed Copy, padding targets have $m_{n,j} = 0$; for Reverse, every position has $m_{n,j} = 1$.

Token accuracy at test length L is defined as

$$A_{\text{tok}}(L) = 100 \frac{\sum_{n=1}^N \sum_j m_{n,j} \mathbb{I}[\hat{y}_{n,j} = y_{n,j}]}{\sum_{n=1}^N \sum_j m_{n,j}}.$$

It measures the percentage of evaluated target tokens predicted correctly. Padding positions do not contribute to either the numerator or denominator.

Exact-match accuracy is defined as

$$A_{\text{EM}}(L) = \frac{100}{N} \sum_{n=1}^N \mathbb{I}[\hat{y}_{n,j} = y_{n,j} \text{ for every position satisfying } m_{n,j} = 1].$$

A sequence is therefore counted as correct only when every non-padding target token is predicted correctly. Exact-match accuracy was treated as the primary synthetic-task metric because each task

requires a complete output sequence. Token accuracy was retained as a complementary metric because it distinguishes complete sequence failure from outputs containing some correct tokens.

For interpretability, both metrics were also reported as error percentages:

$$E_{\text{tok}}(L) = 100 - A_{\text{tok}}(L), \quad E_{\text{EM}}(L) = 100 - A_{\text{EM}}(L).$$

Token error was compared with a uniform-prediction reference over the active target vocabulary. This reference is 90one of ten digits at active positions, and 99Reverse, which use 100 possible content symbols. These references apply to token error and should not be confused with the 10threshold defined below.

3.8.1 Multi-seed aggregation

For every task, positional encoding, and test length, performance was aggregated across the three seeds. If the seed-level measurements are A_1, A_2, A_3 , their mean is

$$\bar{A} = \frac{1}{3} \sum_{s=1}^3 A_s.$$

The sample standard deviation was used to describe variation between seeds, while the standard error was calculated as

$$\text{SE} = \frac{\text{SD}}{\sqrt{3}}.$$

Standard deviation and standard error were not treated as interchangeable. Standard deviation describes the observed dispersion between the three runs, whereas standard error describes uncertainty in their estimated mean. Every result figure specifies explicitly which quantity, if any, is displayed by its error bars or shaded region.

3.8.2 Baseline eligibility

Baseline eligibility was determined before extrapolation performance was interpreted. A seed was considered to have fitted the training task when

$$A_{\text{EM}}(L_{\text{train}}) \geq 90\%.$$

A task-encoding configuration was classified as *eligible* only when all three seeds met this threshold. A configuration was classified as *seed unstable* when the threshold was met by only one or two seeds. Configurations for which no seed met the threshold were classified as *not eligible*. Only eligible task-encoding configurations were used as the principal baselines for conclusions about extrapolation failure.

The 90% criterion is an operational eligibility threshold rather than a claim that 90% constitutes perfect learning. Its purpose is to prevent training-length underfit or seed instability from being misreported as length-generalisation failure.

3.9 Performance Degradation Curves

Performance degradation curves plot accuracy or error percentage against the nominal test length. They show the training-length result together with every longer evaluation point, making it possible to distinguish an immediate collapse at the first unseen length from degradation that emerges only at larger lengths.

For each eligible task-encoding configuration, the first collapse length was defined using the mean exact-match accuracy across seeds:

$$L_{\text{collapse}} = \min \{L > L_{\text{train}} : \bar{A}_{\text{EM}}(L) < 10\%\}.$$

If mean exact-match accuracy fell below 10% at the first unseen length, the observed behaviour was classified as *immediate extrapolation collapse*. If the threshold was crossed only at a later length, the behaviour was classified as delayed degradation followed by collapse. If no evaluated length fell below the threshold, the result was described as *no observed collapse within the evaluated range*.

The 10% value is an operational reporting threshold. It is not the theoretical chance accuracy of an entire output sequence and does not, by itself, identify the internal cause of failure. Its purpose is to

provide a consistent criterion for locating severe degradation across tasks whose complete output lengths differ.

The eligible-baseline plots show the multi-seed mean together with one standard deviation and faint individual-seed curves. This presentation makes both the average length trend and seed-level stability visible. Training length is marked using a vertical reference line, and the 10% exact-match threshold is marked using a horizontal reference line.

Because the evaluated lengths increase approximately geometrically, they are displayed at equally spaced horizontal positions with their true numerical values retained as tick labels. This improves label readability and prevents the shorter lengths from being compressed near the left edge. The equal visual spacing should therefore not be interpreted as a linear numerical distance between consecutive sequence lengths.

3.10 Attention Diagnostics

Accuracy and error percentages identify performance changes but do not directly describe how the model’s attention distributions change with sequence length. Attention statistics were therefore collected as supporting diagnostics for the eligible baseline configurations. Because attention weights are not necessarily faithful explanations of a model’s prediction [8], these statistics were not treated as a complete mechanistic account of failure.

Attention was examined for the five eligible task–encoding configurations: Addition with learned and sinusoidal encodings, Delayed Copy with sinusoidal and RoPE encodings, and Reverse with learned positional embeddings. The analysis used the seed-42 checkpoint as a representative run. For each configuration, attention was extracted at the training length, the first unseen length, and the longest evaluated length.

For an attention distribution α_{ij} , where i is the query position and j is the key position, three statistics were calculated. Normalised attention entropy was defined as

$$H_{\text{norm}} = \frac{-\sum_{j=1}^S \alpha_{ij} \log(\alpha_{ij} + \epsilon)}{\log S},$$

where S is the encoded sequence length and $\epsilon = 10^{-9}$ provides numerical stability. Values near zero indicate concentrated attention, while values closer to one indicate a more diffuse distribution. The statistic was averaged across query positions, attention heads, examples, and Transformer layers.

Average attention distance was defined as

$$D_{\text{attn}} = \sum_{j=1}^S \alpha_{ij} |i - j|,$$

and was averaged over query positions, heads, examples, and layers. This measures the expected absolute positional separation between a query and the positions receiving its attention.

Finally, the local attention ratio was defined as

$$R_{\text{local}} = \sum_{j: |i-j| \leq 10} \alpha_{ij}.$$

It measures the proportion of attention assigned within a fixed window of ten positions on either side of each query. The window was held constant at every evaluation length.

Attention was collected from the first five test batches at each selected length. Because the baseline batch size was eight for Addition and one for Delayed Copy and Reverse, this corresponded to at most 40 Addition examples and five examples for each of the other tasks. The attention analysis is therefore a representative diagnostic rather than a full multi-seed statistical evaluation.

Average distance and local-attention ratio also depend mechanically on sequence length. A fixed local window represents a decreasing fraction of a longer sequence, while an unnormalised distance has a larger possible range at longer lengths. These measures were consequently interpreted as descriptive changes accompanying failure, not as independent proof of its cause.

3.11 Mitigation Experiments

Mitigation experiments were conducted after the baseline analysis. Each intervention was associated with a specific hypothesis about the observed failure behaviour rather than being treated as a general-purpose solution. The experiments used seed 42 and were compared with the corresponding seed-42 original baseline. They were therefore targeted follow-up experiments, not a second complete multi-seed matrix.

Three principal interventions were examined: mixed-length training, curriculum learning, and randomised padded training. Curriculum learning was motivated by the principle of increasing task difficulty gradually during optimisation [5]. Randomised padded training was motivated by work examining whether variation in positional exposure can improve length generalisation [17]. The implementation used here was a narrower test and should not be interpreted as a direct replication of that work.

Intervention	Applied conditions	Training design	Hypothesis tested
Mixed length	Addition–Learned and Delayed Copy–Sinusoidal	Joint exposure to lengths below and including the original training length	Broader length exposure may reduce dependence on one fixed training length.
Curriculum	Delayed Copy–Sinusoidal, Delayed Copy–RoPE, and Reverse–Learned	Sequential stages at lengths 32, 64, 96, and 128	Gradual increases in length may overcome an optimisation barrier associated with direct training at length 128.
Randomised padded	Delayed Copy–Sinusoidal and Delayed Copy–RoPE	Lengths 32, 64, 96, and 128 combined in one shuffled dataset and right-padded to a common maximum	Variation in separator and target positions may reduce reliance on their fixed absolute locations.
Single-length control	Delayed Copy–Sinusoidal	Original length-128 distribution reproduced through the mitigation pipeline	Any observed difference should result from the intervention rather than the alternative training pipeline.

Table 3.8: Mitigation conditions and the specific hypotheses they were designed to test.

For Addition, the first mixed-length variant distributed 10,000 training examples across operand lengths 4, 8, 12, and 16 and trained for 10 epochs. A strengthened variant used 40,000 examples and 20 epochs to test whether the first condition was limited by data volume or optimisation time.

For Delayed Copy, mixed-length and randomised-padded conditions used lengths 32, 64, 96, and 128 with 40,000 examples in total. In the mixed-length condition, each length was processed through a separate fixed-length data loader. In the randomised-padded condition, all examples were combined, shuffled, and right-padded to the maximum training sequence length. Padding was not placed at random offsets; the intervention varied the source, separator, and output locations through the sampled task length.

Curriculum models were trained sequentially at lengths 32, 64, 96, and 128. Each stage used 10,000 examples and five epochs, and the final checkpoint from one stage initialised the next. Only the final length-128 checkpoint was evaluated over the complete test-length schedule.

The mitigation configurations used a batch size of eight. This differed from the batch size of one used in the original Delayed Copy and Reverse baselines. The single-length Delayed Copy control was therefore included to test whether the modified training pipeline could reproduce the original sinusoidal baseline without changing the length distribution.

Each intervention was first evaluated at the original training length. A mitigation satisfying

$$A_{\text{EM}}(L_{\text{train}}) \geq 90\%$$

was considered to have retained training fit and could therefore produce an interpretable extrapolation comparison. If it fell below this threshold, its longer-length outcome was classified as inconclusive due to training underfit. For interventions that retained training fit, exact-match accuracy at the first unseen and longest test lengths was compared directly with the corresponding original baseline. The phrase *no observed improvement* is used when an eligible intervention did not increase extrapolation performance within the evaluated range.

The Reverse curriculum configuration used a 102-symbol vocabulary, whereas the original Reverse baseline used 100 symbols. Unless this condition is rerun using the baseline vocabulary, it is not an exactly matched baseline-versus-intervention comparison and is treated only as an exploratory curriculum sensitivity test.

3.12 Real-World Extension

The synthetic benchmark provides controlled tests in which the target rule is known exactly. A separate real-world experiment was included to examine whether input-length sensitivity also appears in multivariate time-series classification. It was treated as an exploratory contrast rather than validation of the synthetic failure taxonomy.

The experiment used SelfRegulationSCP2 from the UEA Multivariate Time Series Classification Archive [3]. The dataset contains EEG recordings from a binary cursor-control task. It provides 200 training examples and 180 test examples, with seven channels and 1152 time points per example. The predefined archive split was retained.

Each channel was standardised using its mean and standard deviation calculated from the training split across examples and time points. The same training-derived statistics were then applied to the test split. No information from the test set was used to calculate the normalisation parameters.

The classifier was trained using the first 256 time points of every training series. Evaluation used prefixes of lengths 256, 512, 1024, and 1152 from the predefined test examples. Unlike the synthetic protocol, increasing the prefix length introduces additional observed measurements and may therefore add useful information or noise. The experiment measures sensitivity to prefix length rather than deterministic rule extrapolation.

The time-series model replaced the synthetic token-embedding layer with a linear projection from the seven input channels to the 128-dimensional model representation. It then applied the same two Transformer blocks and positional conditions used in the synthetic experiments. The representations were mean-pooled across time and passed to a linear two-class classifier.

Models were trained for 50 epochs using AdamW, a learning rate of 10^{-3} , batch size 16, and cross-entropy loss. The five positional conditions were repeated with seeds 42, 123, and 2024, producing 15 trained classifiers. Classification accuracy was calculated on the 180-example test split at each prefix length. Mean accuracy and standard error were then calculated across the three seeds.

Component	Setting
Dataset	SelfRegulationSCP2
Data type	Seven-channel EEG time series
Classes	2
Training examples	200
Test examples	180
Full sequence length	1152 time points
Training prefix	256 time points
Evaluation prefixes	256, 512, 1024, and 1152
Input projection	7 channels to 128 dimensions
Sequence aggregation	Mean pooling across time
Training epochs	50
Batch size	16
Learning rate	10^{-3}
Positional conditions	Learned, sinusoidal, NoPE, ALiBi, and RoPE
Random seeds	42, 123, and 2024
Evaluation metric	Classification accuracy

Table 3.9: Configuration of the exploratory SelfRegulationSCP2 experiment.

The synthetic 90 were not applied to this experiment because its output is a single noisy class label rather than a deterministic token sequence. Its results were therefore interpreted separately and were not used to determine whether a synthetic failure category had been reproduced.

3.13 Failure Taxonomy Construction

The final methodological stage organised the observed behaviours into a descriptive failure taxonomy. The taxonomy was constructed sequentially from training-length fit, seed stability, extrapolation curves, token-error percentages, position-wise error profiles, task-specific diagnostics, and mitigation comparisons. Attention statistics were retained as secondary context but were not used alone to assign a failure category.

The construction followed five questions:

1. Did the configuration learn the training-length task?
2. Was training-length performance stable across all three seeds?
3. If the baseline was eligible, where did severe extrapolation degradation first occur?
4. How were token errors distributed across output positions and task-specific conditions?
5. Did an intervention retain training fit and improve performance at unseen lengths relative to its corresponding baseline?

These questions produced the five behavioural categories summarised in Table 3.10. The categories are not all mutually exclusive. For example, an eligible configuration may exhibit immediate extrapolation collapse, have an approximately uniform position-wise error profile, and show no observed improvement under a tested intervention.

Behavioural category	Operational criterion	Interpretive role
Training-length underfitting	The configuration does not reach 90% exact match at the training length	Prevents failure to learn the original task from being misreported as extrapolation failure.
Seed instability	The 90% training-fit threshold is met by some, but not all, seeds	Identifies conclusions that depend on data sampling, initialisation, or stochastic optimisation.
Immediate extrapolation collapse	An eligible configuration has mean exact-match accuracy below 10% at the first unseen length	Locates severe degradation directly beyond the observed training-length boundary without assigning a specific internal cause.
Spatial error structure	Token errors are described by their distribution across relative output positions, carry conditions, or dependency distances	Distinguishes approximately uniform near-chance collapse from more position-dependent failure behaviour.
No observed improvement under tested mitigation	An intervention retains at least 90% training-length exact match but does not increase performance at the evaluated unseen lengths relative to its baseline	Reports the outcome of the specific intervention without claiming that the failure is generally resistant to mitigation.

Table 3.10: Operational construction of the failure taxonomy.

Mitigation conditions that failed to retain training fit were classified as inconclusive due to underfitting rather than as unsuccessful evidence about extrapolation. Similarly, changes in attention entropy, distance, or locality were treated only as accompanying descriptive observations. This prevented the taxonomy from making stronger mechanistic claims than were supported by the experimental evidence.

The taxonomy is therefore an evidence-organising framework rather than a statistical proof that each category corresponds to a unique internal model mechanism. Its purpose is to separate observably different outcomes that would otherwise be obscured by reporting only accuracy at the longest test length.

3.14 Output Organisation

Experimental artefacts were separated into configurations, checkpoints, logs, raw result files, derived analyses, and report-ready figures. File names encode the task or dataset, training length, positional condition, and random seed where applicable. This makes individual results traceable to the configuration and checkpoint that produced them.

Raw synthetic baseline results are stored in `outputs/results/multiseed_results/`, mitigation results in `outputs/results/mitigation_results/`, and time-series results in `outputs/results/realworld/`. These CSV files contain the unaggregated measurements for individual task–encoding–seed combinations.

Derived attention statistics are stored in `outputs/analysis/attention/` and `outputs/analysis/attention_summary/`. Position-wise diagnostic CSV files transferred from the GPU environment are retained under `position_diagnostics.aws/final/`. Mitigation and taxonomy summaries are stored under `outputs/analysis/`.

Report-ready PDF and PNG figures, together with the summary CSV files used to construct them, are stored in `outputs/plots/final/`. The PDF versions were used in the dissertation to preserve vector text and line rendering, while PNG versions were retained for rapid inspection.

Raw CSV files were retained separately from aggregated summaries so that reported means, error percentages, eligibility decisions, and plots could be checked against their underlying seed-level measurements. Plotting and analysis scripts read these stored results rather than manually transcribed values.

Artefact	Location
Experiment configurations	configs/multiseed/, configs/mitigation/, and configs/realworld/
Model checkpoints	outputs/checkpoints/
Execution logs	outputs/logs/
Raw baseline results	outputs/results/multiseed_results/
Raw mitigation results	outputs/results/mitigation_results/
Raw real-world results	outputs/results/realworld/
Attention diagnostics	outputs/analysis/attention/ and outputs/analysis/attention_summary/
Position-wise diagnostics	position_diagnostics_aws/final/
Final figures and report summaries	outputs/plots/final/

Table 3.11: Organisation of experimental artefacts and derived report outputs.

3.15 Implementation and Compute Setup

The experimental pipeline was implemented in Python using PyTorch [14]. NumPy was used for numerical preprocessing, pandas for result aggregation, Matplotlib for figure generation, PyYAML for configuration loading, and aeon for access to the UEA time-series dataset.

Synthetic task generators were implemented as separate Python classes. Generated examples were stored in JSON Lines format, with one input–target pair per record. Experimental settings were stored in YAML configuration files rather than embedded directly in training commands. Training, evaluation, diagnostic analysis, and figure generation were implemented as separate executable modules.

Project component	Repository location
Synthetic task generation	src/tasks/
Dataset loading and preprocessing	src/data/
Transformer and positional implementations	src/models/
Training, evaluation, and analysis scripts	src/training/
Baseline configurations	configs/multiseed/
Mitigation configurations	configs/mitigation/
Real-world configurations	configs/realworld/
Dependency specification	requirements.txt
Project instructions and commands	README.md

Table 3.12: Source-code and configuration components supporting reproducibility.

Local macOS execution was used for implementation, debugging, result aggregation, and figure generation. GPU-intensive training and diagnostic evaluation were performed on an Amazon Web Services G5-family instance using CUDA. The same Git repository and YAML-controlled experiment interface were used in both environments.

Random seeds were recorded in every experimental configuration, while training and evaluation logs preserved the executed condition and output location. Checkpoints and raw CSV results were transferred from the GPU environment before the final summaries and figures were generated locally.

The repository targets Python 3.11 and records its required packages in `requirements.txt`. Because that file lists dependency names without exact version pins, complete byte-for-byte environment reconstruction is not guaranteed. This is a reproducibility limitation; the configurations, random seeds, raw results, checkpoints, and analysis scripts nevertheless preserve the principal experimental workflow and reported measurements.

Chapter 4

Synthetic Benchmark Results

This chapter reports the results of the controlled synthetic benchmark. The analysis first establishes which task–encoding configurations learned the training-length task reliably across all three seeds. Extrapolation behaviour is then reported only for these eligible baselines, while configurations that underfit or exhibited seed instability are analysed separately.

Five of the fifteen task–encoding configurations satisfied the multi-seed baseline-eligibility criterion. Every one of these eligible baselines fell below the 10% exact-match threshold at the first unseen length. The principal result is therefore not simply that some models performed poorly at long lengths, but that configurations with stable training-length fit underwent immediate and reproducible extrapolation collapse when the nominal task length first exceeded the training boundary.

The chapter first reports training fit and seed stability across the complete baseline matrix. It then examines eligible-baseline extrapolation and error percentages for Addition, Delayed Copy, and Reverse. Position-wise and task-specific error structure is analysed separately in Chapter 5.

4.1 Overview of Synthetic Results

The baseline matrix contained fifteen task–encoding configurations, each evaluated using seeds 42, 123, and 2024. A task–encoding configuration was eligible for the principal extrapolation analysis only when all three seeds achieved at least 90% exact-match accuracy at the training length.

Figure 4.1 shows the training-length exact-match accuracy of every seed. The point shape identifies the seed, the horizontal black segment gives the mean, and the dashed line marks the 90% eligibility threshold.

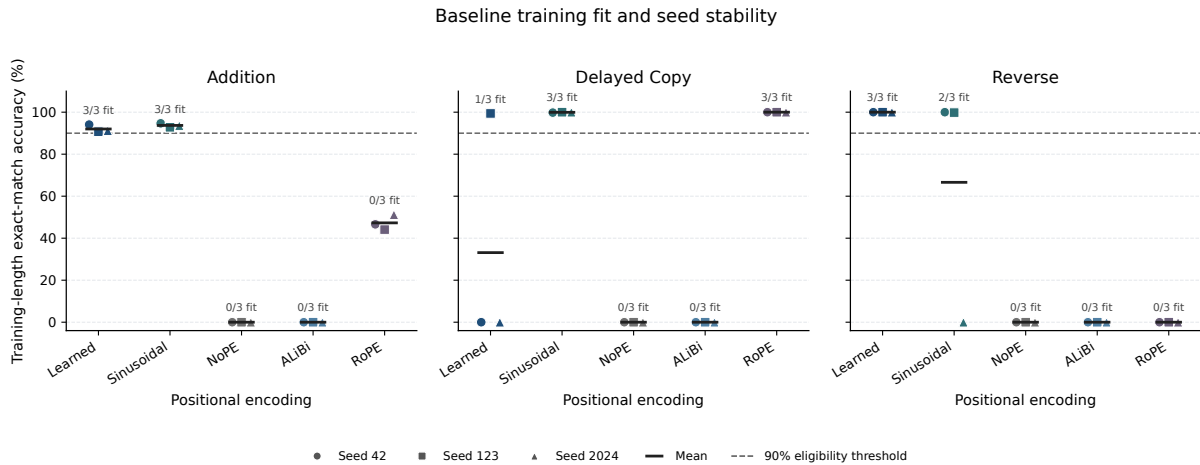


Figure 4.1: Training-length exact-match accuracy across tasks, positional encodings, and seeds. Circles, squares, and triangles represent seeds 42, 123, and 2024 respectively; horizontal black segments show the three-seed mean. The dashed line marks the 90% baseline-eligibility threshold, and annotations report the number of seeds satisfying it.

Five configurations provided stable eligible baselines: learned and sinusoidal encodings for Addition, sinusoidal and RoPE encodings for Delayed Copy, and learned positional embeddings for Reverse. Delayed Copy with learned embeddings was seed unstable because only seed 123 learned the training task. Reverse with sinusoidal encoding was also seed unstable because seeds 42 and 123 succeeded while seed 2024 failed.

The remaining configurations did not satisfy the eligibility criterion. NoPE and ALiBi failed to learn the training task across all three tasks. Addition with RoPE achieved partial training fit but remained below the eligibility threshold for every seed, while Reverse with RoPE also failed at the training length. These configurations cannot support conclusions about length extrapolation because the original task was not first learned reliably.

The eligible configurations showed a common sequence-level outcome: mean exact-match accuracy fell to 0% at the first unseen length and remained at 0% throughout the evaluated range. Token-level results, however, differed between Addition and the two 100-symbol tasks. The following sections therefore report both sequence and token error rather than describing every 0% exact-match result as the same form of failure.

4.2 Addition Results

Addition models were trained using operands of length 16 and evaluated using operand lengths from 16 to 1024. Learned and sinusoidal positional encodings were the only Addition configurations that satisfied the multi-seed eligibility criterion.

At the training length, learned positional embeddings achieved a mean exact-match accuracy of 92.0%, with a standard deviation of 1.84 percentage points and a seed range of 90.7–94.1%. Sinusoidal encoding achieved a mean of 93.7%, with a standard deviation of 0.95 percentage points and a range of 92.8–94.7%. All six corresponding seed-level runs therefore exceeded the 90% eligibility threshold.

Figure 4.2 shows the extrapolation curves for these two eligible baselines. Both fell to 0% exact-match accuracy at length 32, the first unseen length, and remained at 0% through length 1024. The collapse was reproduced across all three seeds rather than being caused by one unstable run.

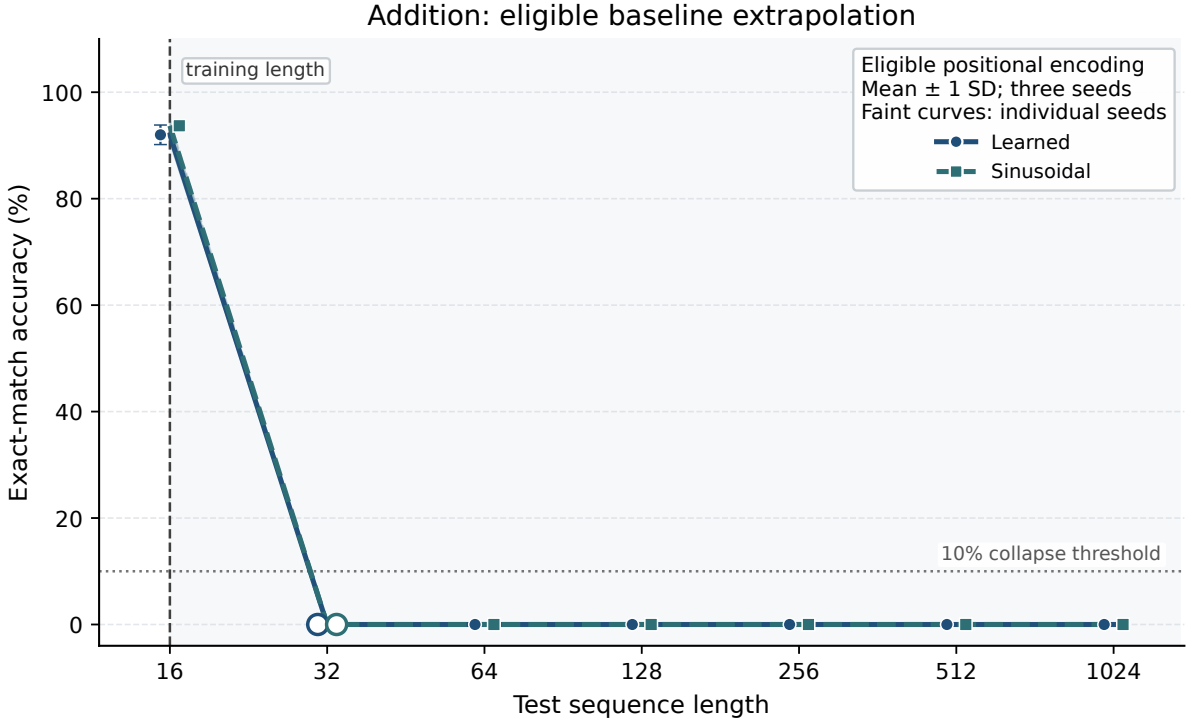


Figure 4.2: Exact-match accuracy of the eligible Addition baselines across operand lengths. Bold curves show the three-seed mean, variability is shown as mean $\pm$ one standard deviation, and faint curves show individual seeds. The vertical dashed line marks the training length and the horizontal dotted line marks the 10% operational collapse threshold.

This result is classified as immediate extrapolation collapse: both configurations learned the length-16

task consistently, but their mean exact-match accuracy crossed the collapse threshold at the first unseen length. The accuracy curve establishes the location and severity of the failure, but it does not by itself identify the internal strategy learned by the models.

The non-eligible Addition conditions exhibited different training-length behaviour. NoPE and ALiBi achieved 0% mean exact-match accuracy at length 16. RoPE achieved a mean of 47.3%, with a standard deviation of 3.60 percentage points and a range of 44.1–51.2%. RoPE therefore demonstrated partial training fit but did not provide a valid multi-seed extrapolation baseline.

Exact-match accuracy alone does not show whether the failed outputs contained mostly correct digits or were incorrect throughout. Figure 4.3 therefore presents sequence and token error percentages for the eligible configurations.

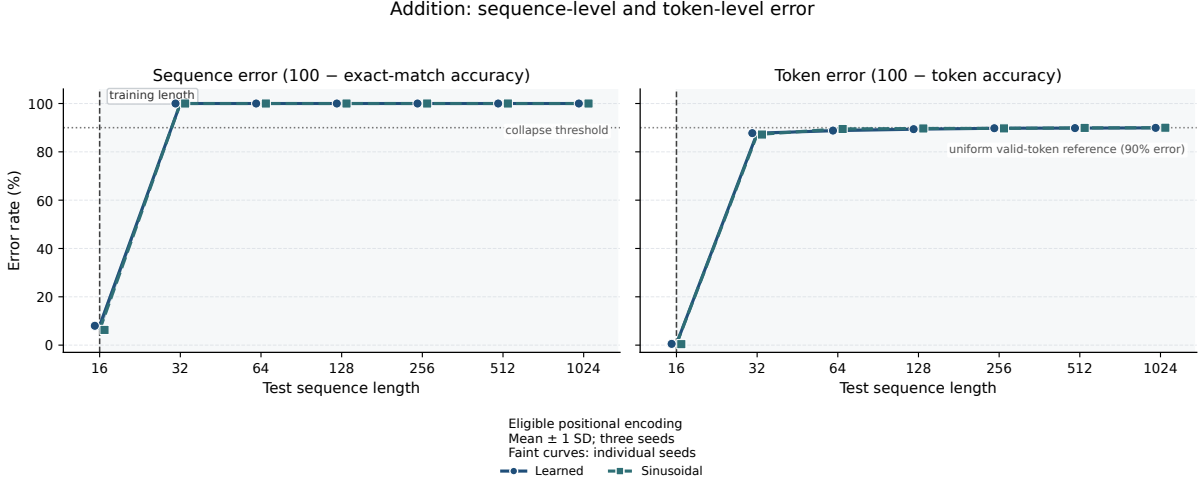


Figure 4.3: Sequence-level and token-level error for the eligible Addition baselines. Sequence error is $100 - A_{\text{EM}}$, while token error is $100 - A_{\text{tok}}$. Bold curves show the three-seed mean, variability is shown as mean $\pm$ one standard deviation, and faint curves show individual seeds. The token-error panel includes the 90% error reference corresponding to uniform prediction over the ten active digit symbols.

At the training length, mean sequence error was 8.0% for learned embeddings and 6.3% for sinusoidal encoding. Their corresponding token errors were only 0.52% and 0.40%, showing that the remaining sequence errors were associated with a small proportion of incorrect digits.

At length 32, sequence error increased to 100% for both eligible configurations. Token error increased to 87.74% for learned embeddings and 87.16% for sinusoidal encoding. Token error then approached the 90% uniform-prediction reference as length increased, reaching 89.93% and 89.98% respectively at length 1024.

Therefore, the 0% exact-match result was not produced by otherwise near-correct sums containing only one occasional error. At unseen lengths, the majority of active output digits were incorrect, and performance moved towards the uniform digit-prediction reference. The spatial distribution of these errors and their relationship to incoming carry are examined in Chapter 5.

4.3 Delayed Copy Results

As specified in Section 3.2, Delayed Copy requires the model to reproduce each content token after a separator and a delay. Models were trained at a nominal length of 128 and evaluated at lengths 256, 512, and 1024. The encoded sequence length was $2L + 1$, so increasing the nominal length also increased the positional separation between each source token and its corresponding target position.

Sinusoidal encoding and RoPE were the two eligible Delayed Copy baselines. Sinusoidal encoding achieved a mean training-length exact-match accuracy of 99.93%, with a standard deviation of 0.12 percentage points and a seed range of 99.8–100.0%. RoPE achieved 100% exact-match accuracy for all three seeds.

Figure 4.4 shows that both configurations fell to 0% exact-match accuracy at length 256, the first unseen length. They remained at 0% at lengths 512 and 1024. The result was reproduced across all three seeds for both positional encodings.

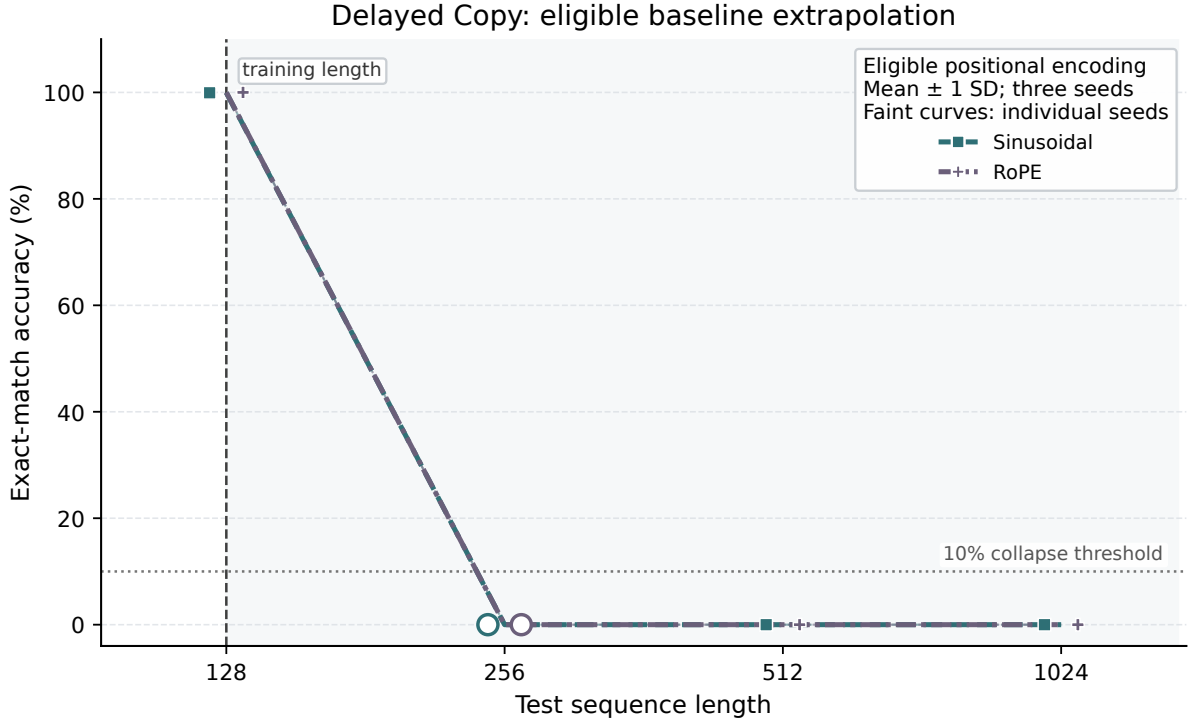


Figure 4.4: Exact-match accuracy of the eligible Delayed Copy baselines across nominal sequence lengths. Bold curves show the three-seed mean, variability is shown as mean $\pm$ one standard deviation, and faint curves show individual seeds. The vertical dashed line marks the training length and the horizontal dotted line marks the 10% operational collapse threshold.

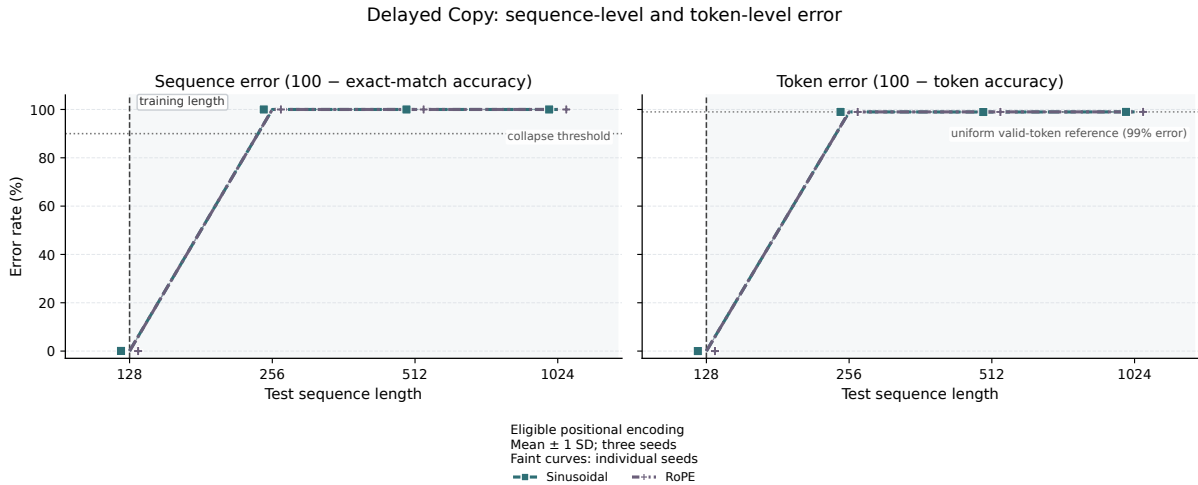


Figure 4.5: Sequence-level and token-level error for the eligible Delayed Copy baselines. Bold curves show the three-seed mean, variability is shown as mean $\pm$ one standard deviation, and faint curves show individual seeds. The token-error panel includes the 99% error reference corresponding to uniform prediction over the 100 content symbols.

This behaviour is classified as immediate extrapolation collapse. The models reproduced almost every training-length sequence exactly, but neither configuration produced one completely correct sequence among the 3,000 seed-level test examples at length 256. The result locates the failure at the first unseen length without requiring an assumption about the internal retrieval strategy used by the model.

The token-level results show that the failed sequences were not close to correct. Figure 4.5 presents sequence and token error percentages for the eligible configurations.

At the training length, mean sequence error was 0.07% for sinusoidal encoding and 0% for RoPE. Their token errors were approximately 0.0005% and 0%, respectively. At length 256, sequence error increased to 100% for both configurations, while token error increased to 99.00% for sinusoidal encoding and 98.98% for RoPE. Token errors remained close to the 99% uniform-symbol reference at lengths 512 and 1024.

Therefore, the failure was not limited to a small number of misplaced tokens. At unseen lengths, predictions across the active target positions were approximately as inaccurate as uniform prediction over the content vocabulary.

Learned positional embeddings did not provide an eligible pooled baseline. Only seed 123 learned the training task, achieving 99.4% exact-match accuracy, while seeds 42 and 2024 achieved 0%. This produced a mean of 33.13% and a standard deviation of 57.39 percentage points. The configuration is therefore classified as seed unstable, and its pooled mean is not interpreted as a representative extrapolation curve.

NoPE and ALiBi achieved 0% exact-match accuracy at the training length for all three seeds. Their longer-length results are consequently classified as not applicable to extrapolation rather than as additional evidence of length-generalisation failure.

4.4 Reverse Results

Reverse requires each output position to retrieve the input token measured from the opposite end of the sequence. Models were trained at length 128 and evaluated at lengths 256, 512, and 1024. Learned positional embeddings were the only Reverse configuration that satisfied the multi-seed eligibility criterion.

The learned condition achieved 100% exact-match and token accuracy at the training length for all three seeds. Figure 4.6 shows that mean exact-match accuracy then fell to 0% at length 256 and remained at 0% through length 1024. No seed produced a non-zero exact-match result at any unseen length.

This is another case of immediate extrapolation collapse. Unlike a training-underfit configuration, the learned baseline solved every training-length test example across all seeds before failing at the first unseen length. The result demonstrates that stable in-distribution performance did not extend across the change from length 128 to 256.

Figure 4.7 shows that the sequence-level collapse was accompanied by token errors close to the uniform-symbol reference.

Mean token error increased from 0% at the training length to 98.75% at length 256. It remained close to the 99% uniform-symbol reference, reaching 98.84% at length 512 and 98.92% at length 1024. The model therefore did not merely produce reversals containing a small number of local mistakes; most output positions were incorrect once the sequence length exceeded the training boundary.

Sinusoidal encoding was seed unstable. Seeds 42 and 123 achieved 100% training-length exact-match accuracy, while seed 2024 achieved 0%. This produced a pooled mean of 66.6% and a standard deviation of 57.68 percentage points. Pooling these qualitatively different runs would obscure the instability, so sinusoidal encoding was not treated as an eligible three-seed baseline.

NoPE, ALiBi, and RoPE achieved 0% training-length exact-match accuracy for all three seeds. Their results are classified as training-length underfitting and do not support conclusions about extrapolation. The distribution of Reverse errors by relative output position and dependency distance is examined in Chapter 5.

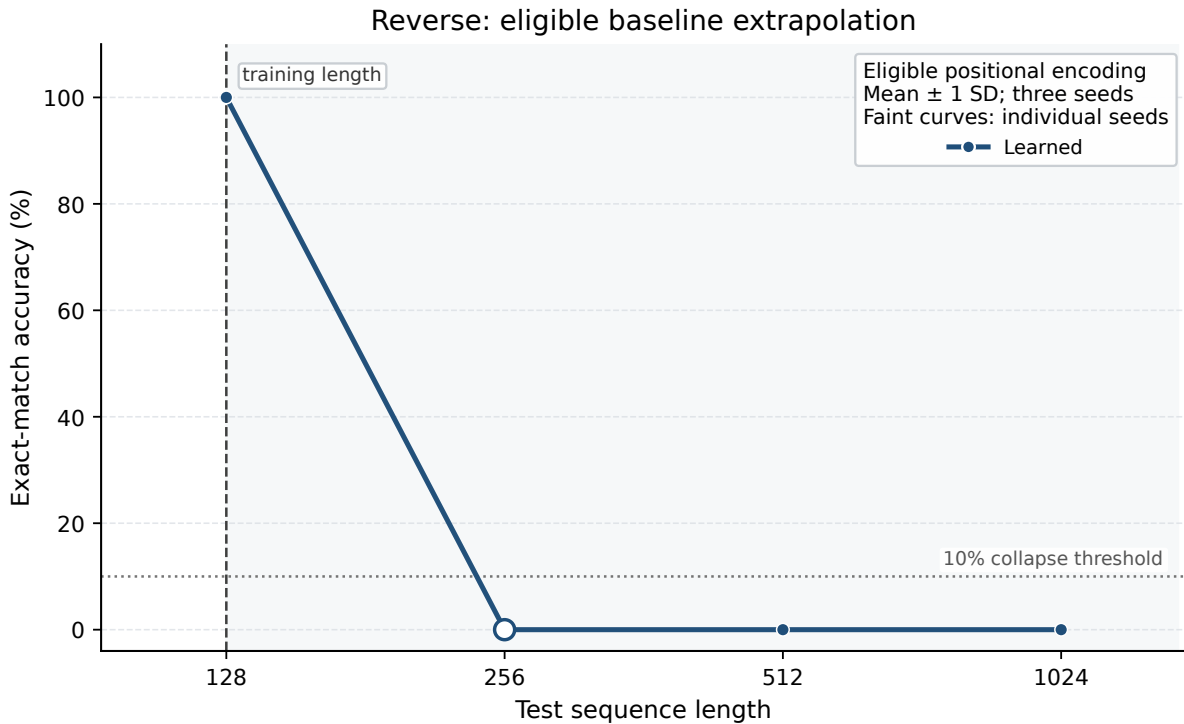


Figure 4.6: Exact-match accuracy of the eligible Reverse baseline across sequence lengths. The bold curve shows the three-seed mean, variability is shown as mean $\pm$ one standard deviation, and faint curves show individual seeds. The vertical dashed line marks the training length and the horizontal dotted line marks the 10% operational collapse threshold.

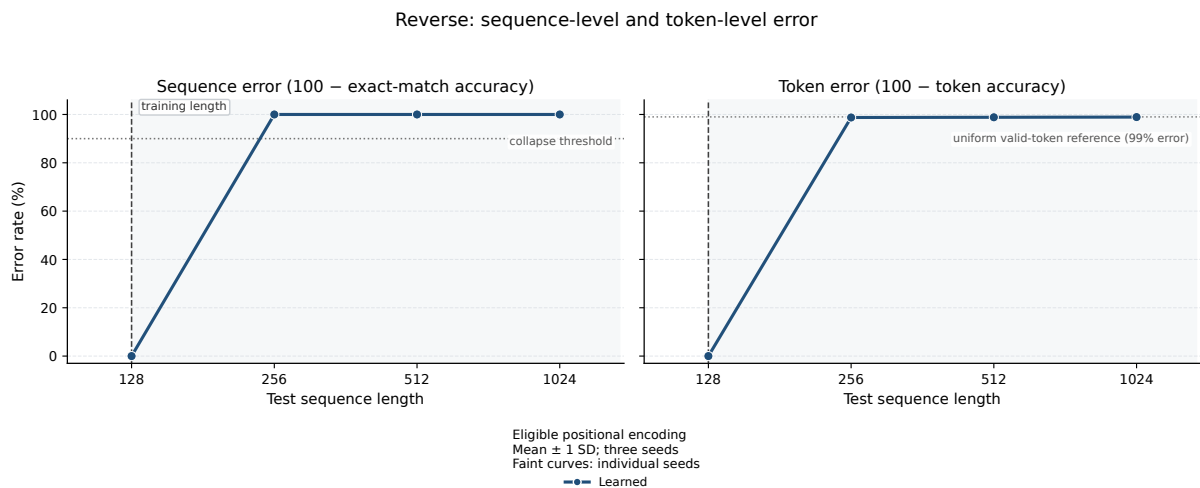


Figure 4.7: Sequence-level and token-level error for the eligible Reverse baseline. The bold curve shows the three-seed mean, variability is shown as mean $\pm$ one standard deviation, and faint curves show individual seeds. The token-error panel includes the 99% error reference corresponding to uniform prediction over the 100-symbol vocabulary.

4.5 Positional Encoding Comparison

The positional-encoding conditions produced substantially different training-length outcomes. However, their effects were task dependent rather than forming a single consistent ranking. Learned positional embeddings were eligible for Addition and Reverse but seed unstable for Delayed Copy. Sinusoidal encoding was eligible for Addition and Delayed Copy but seed unstable for Reverse. RoPE was eligible only for Delayed Copy.

Table 4.1 reports the training-length result for all fifteen task–encoding configurations. The number of successful seeds shows how many runs reached the 90% exact-match eligibility threshold. Training accuracy is reported as the three-seed mean and sample standard deviation.

Task	Encoding	Fit seeds	Exact at train (%)	Baseline interpretation
Addition	Learned	3/3	92.00 $\pm$ 1.84	Eligible; immediate collapse
Addition	Sinusoidal	3/3	93.70 $\pm$ 0.95	Eligible; immediate collapse
Addition	NoPE	0/3	0.00 $\pm$ 0.00	Training-length underfit
Addition	ALiBi	0/3	0.00 $\pm$ 0.00	Training-length underfit
Addition	RoPE	0/3	47.30 $\pm$ 3.60	Partial fit; not eligible
Delayed Copy	Learned	1/3	33.13 $\pm$ 57.39	Seed unstable
Delayed Copy	Sinusoidal	3/3	99.93 $\pm$ 0.12	Eligible; immediate collapse
Delayed Copy	NoPE	0/3	0.00 $\pm$ 0.00	Training-length underfit
Delayed Copy	ALiBi	0/3	0.00 $\pm$ 0.00	Training-length underfit
Delayed Copy	RoPE	3/3	100.00 $\pm$ 0.00	Eligible; immediate collapse
Reverse	Learned	3/3	100.00 $\pm$ 0.00	Eligible; immediate collapse
Reverse	Sinusoidal	2/3	66.60 $\pm$ 57.68	Seed unstable
Reverse	NoPE	0/3	0.00 $\pm$ 0.00	Training-length underfit
Reverse	ALiBi	0/3	0.00 $\pm$ 0.00	Training-length underfit
Reverse	RoPE	0/3	0.00 $\pm$ 0.00	Training-length underfit

Table 4.1: Training-length performance and eligibility across the complete synthetic baseline matrix. Exact-match values are the mean $\pm$ one sample standard deviation over three seeds. A fit seed achieved at least 90% training-length exact-match accuracy.

Five configurations were eligible, two were seed unstable, and eight were not eligible. All five eligible configurations subsequently achieved 0% mean exact-match accuracy at the first unseen length and at the longest evaluated length. Therefore, none of the positional conditions produced reliable sequence-level extrapolation within an eligible task–encoding configuration.

The non-eligible results must be interpreted differently. NoPE and ALiBi failed to meet the training-fit criterion on every task in this particular architecture and optimisation setting. This finding does not establish that these methods generally cannot support length extrapolation; it shows that their present configurations did not provide valid trained baselines from which extrapolation could be assessed.

RoPE also displayed a strongly task-dependent outcome. It achieved 100% training-length exact match across all seeds on Delayed Copy, partial training-length performance on Addition, and 0% exact match on Reverse. The comparison therefore provides no evidence that one positional encoding is uniformly preferable across the three task structures.

Finally, although every configuration had 0% mean exact-match accuracy at the longest test length, this common endpoint does not imply a common failure type. For eligible configurations, it represents extrapolation collapse. For seed-unstable configurations, pooled results conceal different training outcomes. For non-eligible configurations, it primarily reflects failure to establish the training task.

4.6 Failure Length Analysis

The first collapse length was calculated only for configurations that satisfied the multi-seed baseline-eligibility criterion. As defined in Section 3.9, it is the first evaluated unseen length at which mean exact-match accuracy falls below 10%.

Table 4.2 reports the result for all five eligible baselines. Addition crossed the threshold at length 32, while Delayed Copy and Reverse crossed it at length 256. In every case, this was the first evaluated length beyond L_{train} .

All five eligible baselines therefore exhibited immediate extrapolation collapse according to the operational definition. No eligible configuration showed a gradual sequence-level decline across multiple unseen evaluation points before crossing the threshold. Instead, exact-match accuracy changed from at least 92% at the training length to 0% at the first unseen length.

Task	Encoding	L_{train}	First length	unseen	First length	collapse
Addition	Learned	16	32		32	
Addition	Sinusoidal	16	32		32	
Delayed Copy	Sinusoidal	128	256		256	
Delayed Copy	RoPE	128	256		256	
Reverse	Learned	128	256		256	

Table 4.2: First collapse length for every eligible synthetic baseline. Collapse is defined as mean exact-match accuracy below 10% at an unseen length.

The term *immediate* refers to the first sampled unseen length rather than to every possible length directly above the training boundary. Addition was not evaluated at operand lengths between 16 and 32, and Delayed Copy and Reverse were not evaluated at lengths between 128 and 256. The experiment therefore establishes that collapse had occurred by the first evaluated doubling of length, but it cannot identify the precise intermediate length at which the degradation began.

Addition with RoPE is not assigned a collapse length. Although its mean training-length exact-match accuracy was 47.3%, none of its seeds reached the 90% eligibility threshold. Labelling its length-32 result as extrapolation collapse would incorrectly combine partial training fit with failure beyond the training distribution.

The common collapse boundary shows that stable training-length performance did not persist to the first sampled unseen length. It does not, by itself, establish that the models learned representations explicitly tied to one length; determining the internal cause would require stronger mechanistic evidence.

4.7 Multi-Seed Robustness

The multi-seed experiment examined sensitivity to synthetic data sampling, parameter initialisation, minibatch ordering, and stochastic optimisation. Figure 4.1, presented in the chapter overview, shows the complete seed-level training results. It should not be replaced or repeated by a second mean-and-standard-error figure.

The five eligible configurations were stable across seeds: each of their individual runs exceeded the 90% training-fit threshold. However, Delayed Copy with learned positional embeddings and Reverse with sinusoidal encoding produced qualitatively different outcomes across seeds, as shown in Table 4.3.

Task	Encoding	Seed 42	Seed 123	Seed 2024	Fit seeds
Delayed Copy	Learned	0.0%	99.4%	0.0%	1/3
Reverse	Sinusoidal	100.0%	99.8%	0.0%	2/3

Table 4.3: Training-length exact-match accuracy for the two seed-unstable configurations. A fit seed achieved at least 90% exact-match accuracy.

For Delayed Copy with learned embeddings, only seed 123 learned the training-length task. Pooling the three runs produced a mean of 33.13% and a standard deviation of 57.39 percentage points, but the mean does not describe a typical run: it combines one nearly perfect model with two models at 0%.

The same issue occurred for Reverse with sinusoidal encoding. Seeds 42 and 123 achieved 100.0% and 99.8%, respectively, while seed 2024 achieved 0%. The resulting mean of 66.6% and standard deviation of 57.68 percentage points again conceal two distinct outcomes.

These results justify reporting individual seeds alongside aggregated statistics. Mean performance alone would make both configurations appear to have achieved partial training fit, when the seed-level evidence instead shows successful and unsuccessful optimisation outcomes. For this reason, neither configuration was pooled as an eligible baseline.

Seed stability must also be separated from task success. NoPE and ALiBi were stable in the sense that their training-length exact-match accuracy was consistently 0%, but this consistency does not make them valid extrapolation baselines. Robustness therefore required both adequate training performance and agreement across seeds.

4.8 Generalisation Gap

For descriptive purposes, the training-to-longest generalisation gap can be defined as $\bar{AEM}(L_{\max})$.

A large positive value can indicate that performance was high at the training length but low at the longest evaluated length. However, the gap is only meaningful after training fit has been established. A model scoring 0 both lengths has a gap of zero, but this reflects training underfit rather than successful generalisation. In the present benchmark, mean exact-match accuracy at the longest length was 0 configuration’s numerical gap was identical to its training-length mean. The gap therefore added no information beyond the baseline-eligibility analysis and was not used as a primary ranking statistic. Among the eligible baselines, the gaps were 92.0 percentage points for Addition with learned embeddings, 93.7 for Addition with sinusoidal encoding, 99.93 for Delayed Copy with sinusoidal encoding, and 100 for both Delayed Copy with RoPE and Reverse with learned embeddings. These values quantify the difference between the endpoints, but they do not show where failure began. The first-collapse analysis is more informative because it establishes that all five eligible configurations crossed the operational threshold at the first sampled unseen length. The legacy generalisation-gap plot is therefore omitted from the final report. Baseline eligibility, seed-level training fit, and complete degradation curves provide a clearer and less potentially misleading account of the results.

4.9 Comparison with Expected Positional Encoding Behaviour

The results do not support a simple expectation that selecting a positional encoding associated with longer-context evaluation is sufficient to produce algorithmic length generalisation. ALiBi was introduced in a causal language-modelling setting specifically to support evaluation on sequences longer than those used during training [15], while RoPE encodes relative positional relationships through rotations of query and key representations [?]. Previous work has also shown that NoPE can exhibit competitive length-generalisation behaviour on some tasks [9].

In the present experiments, RoPE provided a stable training baseline for Delayed Copy but still collapsed at the first unseen length. It did not provide an eligible baseline for Addition or Reverse. NoPE and the bidirectional ALiBi adaptation did not learn any of the three training tasks under the evaluated architecture and optimisation settings. These outcomes show that the positional mechanism alone did not determine whether the model would first learn the task or subsequently extrapolate.

The results should not be interpreted as a general contradiction of the original ALiBi, RoPE, or NoPE studies. The present model was a compact bidirectional encoder trained from scratch on deterministic sequence-to-sequence tasks. In particular, its ALiBi implementation used a symmetric absolute-distance bias rather than the causal setting evaluated in the original work. The tasks, architecture, optimisation budget, and attention structure therefore differ from those previous evaluations.

Instead, the comparison demonstrates an interaction between task structure and positional treatment. Learned embeddings were eligible for Addition and Reverse but seed unstable for Delayed Copy. Sinusoidal encoding was eligible for Addition and Delayed Copy but seed unstable for Reverse. RoPE was eligible only for Delayed Copy. No positional condition was consistently eligible across all three tasks, and none of the eligible conditions extrapolated at the sequence level.

The benchmark therefore supports the narrower conclusion that positional encoding is one component of length-generalisation behaviour, not a sufficient guarantee of it. A fuller comparison with prior work and the implications for model design are developed in the Discussion chapter.

4.10 Summary of Synthetic Findings

The complete synthetic baseline matrix produced five eligible task-encoding configurations, two seed-unstable configurations, and eight configurations that did not satisfy the training-fit criterion. This separation was necessary because poor performance at an unseen length has a different interpretation depending on whether the model first learned the training task.

All five eligible baselines exhibited immediate extrapolation collapse. Addition with learned and sinusoidal encodings fell from mean training-length exact-match accuracies of 92.0% and 93.7% to 0% at length 32. Delayed Copy with sinusoidal encoding and RoPE fell from 99.93% and 100% to 0% at length 256. Reverse with learned embeddings fell from 100% to 0% at length 256. These outcomes were reproduced across all three seeds.

Token errors established that the sequence-level collapse was severe. Addition token error increased to approximately 87% at the first unseen length and approached its 90% uniform digit-prediction reference at longer lengths. Delayed Copy and Reverse token errors increased to approximately 99%, close to their uniform 100-symbol references. The 0% exact-match results therefore did not generally represent near-correct sequences containing only occasional token errors.

The positional comparison did not identify one encoding that was consistently successful across tasks. It instead revealed task-dependent training fit and two cases of substantial seed instability. These findings reinforce the need to establish a valid multi-seed baseline before interpreting extrapolation performance.

This chapter has established whether failure occurred, its severity in percentage terms, and the first sampled length at which it was observed. The next chapter examines how the errors were distributed across output positions and task-specific conditions, how attention statistics changed with length, and whether the tested training interventions produced any observed improvement.

Chapter 5

Failure Behaviour, Mitigation Strategies, and Taxonomy

The previous chapter established which configurations reliably learned the training task and showed that all five eligible task-encoding baselines collapsed at the first evaluated unseen length. This chapter moves beyond the binary question of whether a model fails and examines how that failure is expressed across the output sequence.

The analysis first examines position-wise token errors and task-specific failure conditions. For Addition, errors are compared across output positions and between digits with and without an incoming carry. For Delayed Copy, errors are examined across the copied output sequence. For Reverse, errors are analysed by output position and dependency distance. Attention statistics are then used as descriptive evidence about changes in attention behaviour across sequence lengths. Finally, the chapter compares mitigation variants with their corresponding original baselines and organises the observed behaviours into a failure taxonomy.

Because the detailed diagnostics use representative seed-42 checkpoints, their purpose is to characterise the form of failure rather than to establish multi-seed prevalence. Multi-seed baseline eligibility and robustness were reported separately in Chapter 4.

5.1 Motivation for Failure Analysis

Exact-match accuracy provides a strict sequence-level measure: an example is counted as correct only when every evaluated output token is predicted correctly. This is appropriate for algorithmic tasks, but once exact-match accuracy reaches zero it does not reveal whether errors are concentrated in particular output regions or distributed across the entire sequence. Aggregate token accuracy provides additional information, but it can still hide systematic variation across positions and task conditions.

The diagnostic analysis therefore uses four complementary sources of evidence. First, position-wise token-error profiles identify whether failure is concentrated near the beginning, middle, or end of the output. Second, task-specific comparisons test whether Addition errors are associated with incoming carries and whether Reverse errors vary with dependency distance. Third, attention statistics describe how the learned attention distributions change as the evaluated sequence length increases. Fourth, matched baseline-mitigation comparisons test whether particular interventions improve performance at unseen lengths while preserving training-length performance.

These sources answer different questions and should not be treated as interchangeable. Position-wise and task-specific diagnostics describe where prediction errors occur. Attention statistics describe associated changes in attention weights, but do not by themselves establish a causal or complete mechanistic explanation of model behaviour [8]. Mitigation comparisons provide intervention evidence, although conclusions remain limited to the particular configurations evaluated.

5.2 Attention Diagnostic Metrics and Scope

Attention diagnostics were computed from saved attention maps for selected seed-42 checkpoints at the training length, the first evaluated unseen length, and the longest test length. As defined in the

methodology chapter, the analysis uses normalised attention entropy, average attention distance, and local attention ratio.

Normalised attention entropy indicates whether attention becomes more concentrated or diffuse relative to the maximum entropy possible at a given sequence length. Average attention distance summarises the distance between query positions and the positions receiving attention. Local attention ratio measures the proportion of attention assigned within a fixed neighbourhood of each query position.

The statistics were averaged over attention heads, layers, and the examples included in the diagnostic batches. The analysis used the first five evaluation batches, corresponding to at most 40 Addition examples and five Delayed Copy or Reverse examples. Consequently, these measurements are treated as descriptive case studies rather than population-level estimates or direct explanations of the prediction failures.

The position-wise and task-specific error analyses are presented before the attention results because they provide the most direct evidence about the observable form of failure.

5.3 Position-Wise Failure Behaviour

The aggregate results establish that exact-match accuracy collapses at the first evaluated unseen length, but they do not reveal whether token errors are concentrated in particular parts of the output. To examine this, valid target positions were divided into ten bins according to their relative position in the output sequence. Token-error rates were then calculated within each bin.

The analysis uses representative seed-42 checkpoints from the five task-encoding configurations that satisfied the multi-seed baseline eligibility criterion. The dotted reference lines indicate the expected token-error rate under uniform prediction over the valid target vocabulary: 90% for the ten Addition digits and 99% for the 100-token Delayed Copy and Reverse vocabularies.

5.3.1 Addition

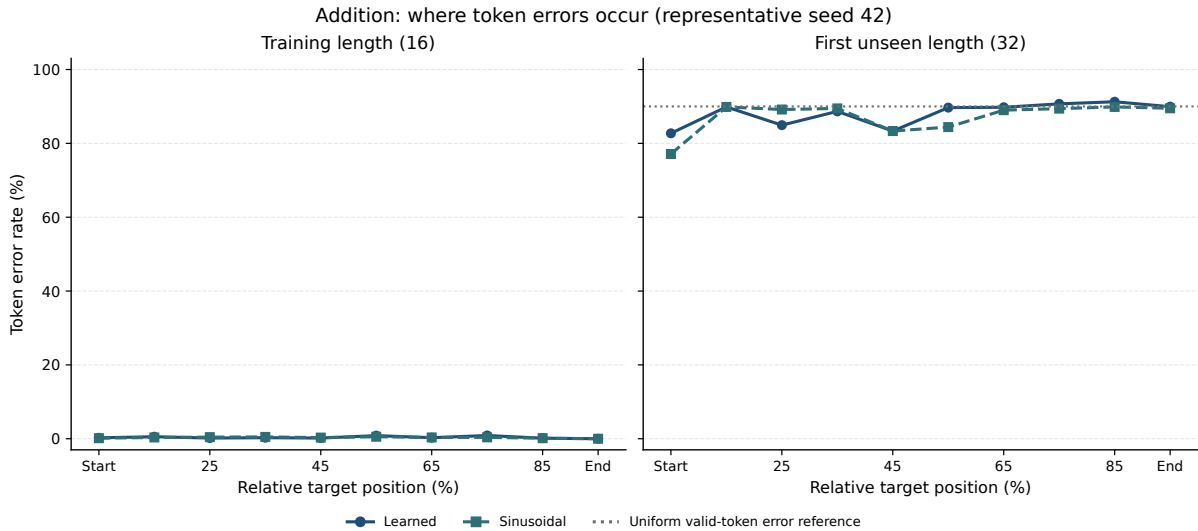


Figure 5.1: Position-wise token-error rates for the eligible Addition configurations at the training length and first evaluated unseen length. Target positions are grouped into ten relative-position bins. The dotted line denotes the 90% uniform digit-prediction error reference. Results use representative seed-42 checkpoints.

Figure 5.1 shows that both Addition models have low token-error rates across the output at the training length. At length 32, errors become distributed throughout the output, but they are not completely uniform across position.

For the learned model, the overall token-error rate at length 32 was 88.0%. The error rate increased from 82.7% in the start region to 86.5% in the middle and 89.9% near the end, giving a positional range of 8.5 percentage points. The sinusoidal model showed a stronger gradient: its error rate increased from 77.1% at the start to 83.9% in the middle and 89.5% at the end, a range of 12.7 percentage points.

The failure is therefore broadly distributed rather than confined to a single output region, but the lower error near the beginning indicates that some limited positional structure remains at the first unseen length. Performance approaches the uniform-prediction reference towards later output positions.

5.3.2 Delayed Copy

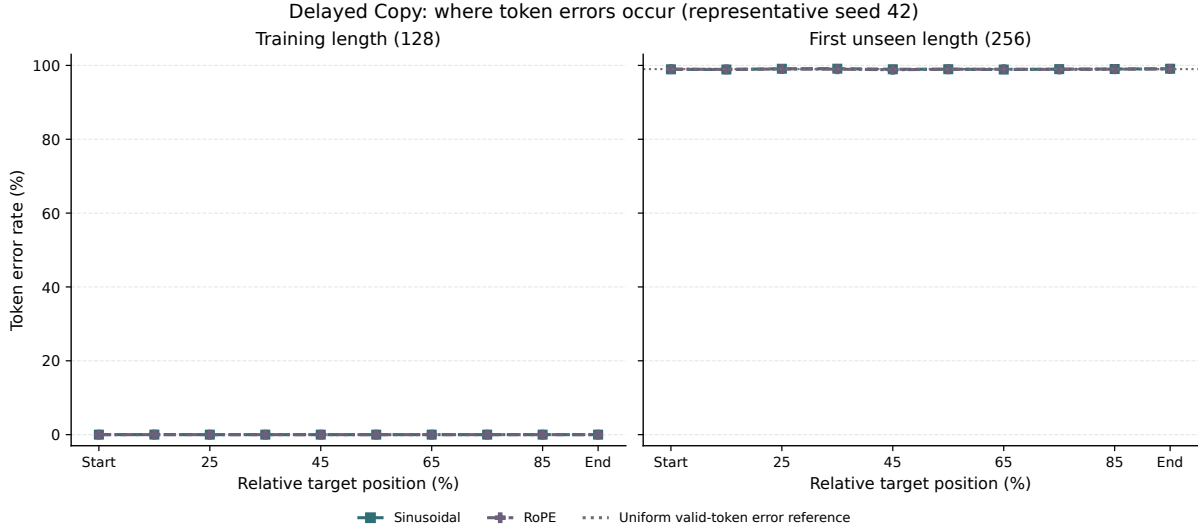


Figure 5.2: Position-wise token-error rates for the eligible Delayed Copy configurations at the training length and first evaluated unseen length. Target positions are grouped into ten relative-position bins. The dotted line denotes the 99% uniform valid-token error reference. Results use representative seed-42 checkpoints.

Figure 5.2 shows a different failure pattern. Sinusoidal and RoPE have approximately zero token error throughout the output at the training length. At length 256, however, both models produce error rates close to the 99% uniform-prediction reference across every relative position.

The overall token-error rates were 99.0% for both configurations. For the sinusoidal model, the start-, middle-, and end-region error rates were 98.94%, 98.97%, and 99.10%, respectively. For RoPE, the corresponding values were 99.00%, 98.90%, and 99.05%. The positional ranges were only approximately 0.23 and 0.22 percentage points.

Delayed Copy therefore exhibits an almost spatially uniform near-reference collapse. The model does not retain a clearly better-performing region near the beginning or end of the copied output when the sequence length doubles.

5.3.3 Reverse

As shown in Figure 5.3, the learned Reverse model has zero token error across the output at the training length. At length 256, the overall token-error rate rises to 98.79%, and errors occur throughout the output sequence.

The start-, middle-, and end-region error rates were 98.87%, 98.85%, and 98.30%, respectively. Their range was only 0.71 percentage points. Consequently, Reverse failure is not concentrated at a particular relative output position. Instead, the model changes from uniformly correct behaviour at the training length to an almost uniformly incorrect output at the first unseen length.

5.3.4 Comparison Across Tasks

The position-wise diagnostics distinguish two forms of spatial behaviour. Addition retains a modest positional gradient at the first unseen length, with lower errors near the beginning and errors approaching the uniform-prediction reference towards the end. Delayed Copy and Reverse show much more uniform failure, with error rates close to their 99% reference throughout the output.

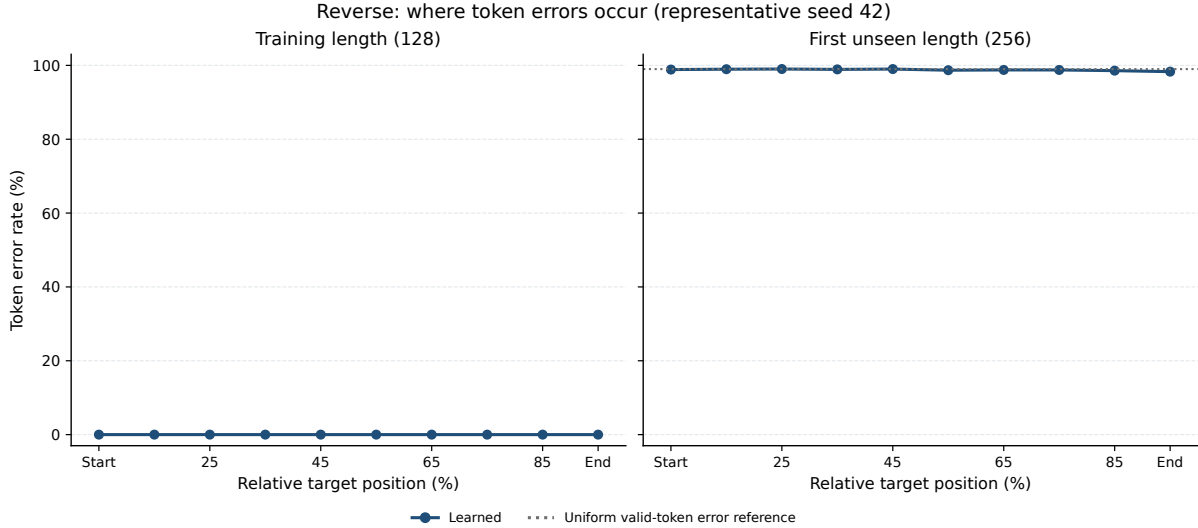


Figure 5.3: Position-wise token-error rates for Reverse with learned positional embeddings at the training length and first evaluated unseen length. Target positions are grouped into ten relative-position bins. The dotted line denotes the 99% uniform valid-token error reference. Results use the representative seed-42 checkpoint.

These results demonstrate why exact-match accuracy alone is insufficient. Although all five eligible configurations have 0% exact-match accuracy at the first unseen length, the distribution of their token errors is not identical.

5.4 Task-Specific Failure Conditions

Position-wise profiles identify where errors occur in the output, but they do not test whether those errors correspond to the computation required by each task. Two additional analyses were therefore performed. Addition errors were conditioned on the presence of an incoming carry, while Reverse errors were conditioned on the distance between each output position and its corresponding source position.

5.4.1 Addition: Incoming-Carry Condition

Figure 5.4 compares ordinary target digits according to whether their calculation receives a carry from the preceding digit column. At the first unseen length, the learned model produced an error rate of 88.14% without an incoming carry and 89.09% with an incoming carry. The difference was 0.96 percentage points. For the sinusoidal model, the corresponding error rates were 88.46% and 88.46%, differing by less than 0.01 percentage points.

The two conditions remain closely aligned at the longer evaluation lengths. Therefore, the observed Addition collapse is not primarily concentrated on digits requiring an incoming carry. This does not show that the models learned a length-general carry procedure; rather, it shows that once extrapolation fails, ordinary digits with and without incoming carries are both predicted poorly.

5.4.2 Reverse: Dependency Distance

For an output position i in a sequence of length L , the corresponding source position in the Reverse task is $L - 1 - i$. The normalised dependency distance used in this analysis is therefore

$$d_i = \frac{|(L - 1 - i) - i|}{L - 1}.$$

A value near zero represents a relatively short input-output dependency, whereas a value near one represents a dependency spanning almost the full sequence.

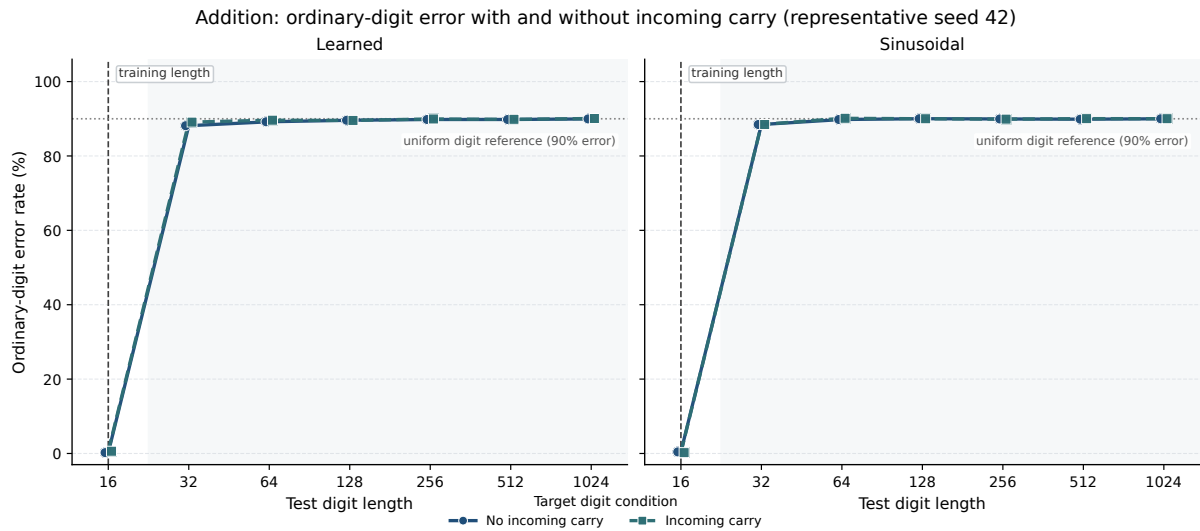


Figure 5.4: Addition error rates for ordinary target digits with and without an incoming carry. The learned and sinusoidal seed-42 checkpoints are shown across all evaluated lengths. The optional leading result digit is excluded from this comparison. The dotted line denotes the 90% uniform digit-prediction error reference.

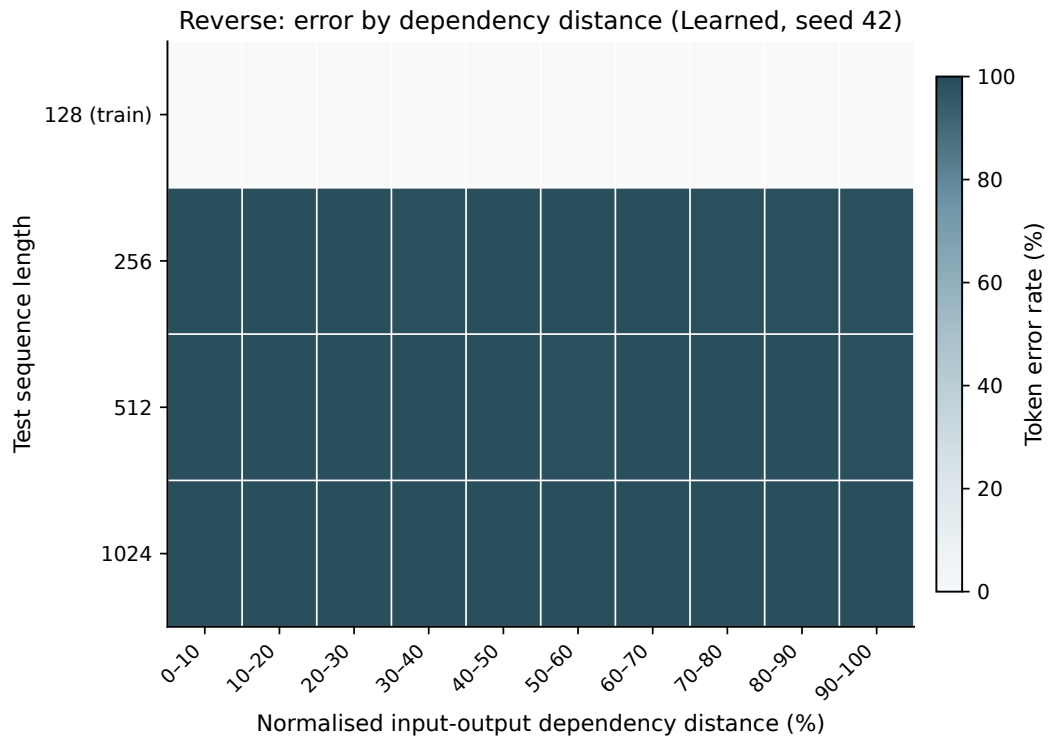


Figure 5.5: Reverse token-error rates grouped by normalised input-output dependency distance for the representative learned seed-42 checkpoint. Distances are divided into ten equal-width bins.

Figure 5.5 shows zero error across all dependency-distance bins at the training length. At the first unseen length, error rates range only from approximately 98.54% to 98.92% across the ten bins. Similarly high and nearly uniform error rates occur at lengths 512 and 1024.

The failure is therefore not concentrated only on output tokens requiring the longest dependencies. Shorter and longer reverse mappings fail at comparable rates once the model is evaluated beyond its training length. This supports the interpretation of a distributed reversal failure rather than a gradual loss of accuracy as dependency distance increases.

5.5 Attention Behaviour Across Tasks

Attention statistics provide descriptive evidence about changes in model behaviour across sequence lengths. Table 5.1 reports normalised attention entropy and local attention ratio for the representative seed-42 checkpoints. Values are descriptive means across the two Transformer layers and the first five diagnostic batches; they are not multi-seed estimates.

Task	Encoding	Normalised entropy			Local attention ratio		
		Train	First OOD	Longest	Train	First OOD	Longest
Addition	Learned	0.157	0.278	0.594	0.496	0.280	0.012
Addition	Sinusoidal	0.345	0.475	0.588	0.534	0.360	0.012
Delayed Copy	Sinusoidal	0.645	0.627	0.630	0.075	0.032	0.007
Delayed Copy	RoPE	0.605	0.577	0.638	0.069	0.027	0.005
Reverse	Learned	0.539	0.731	0.910	0.119	0.073	0.022

Table 5.1: Attention statistics for representative seed-42 checkpoints at the training length, first evaluated out-of-distribution (OOD) length, and longest test length. The fixed local neighbourhood is defined by $|i - j| \leq 10$. Values are descriptive means over two layers and five diagnostic batches.

5.5.1 Addition

For Addition, normalised entropy increased with sequence length for both eligible encodings. For the learned model, it increased from 0.157 at length 16 to 0.278 at length 32 and 0.594 at length 1024. The sinusoidal model showed a similar increase from 0.345 to 0.475 and then 0.588. At the same time, the local attention ratio decreased from approximately 0.50 at the training length to approximately 0.01 at the longest length.

The first entropy increase coincided with exact-match accuracy falling from 94.1% to 0% for the learned seed-42 model and from 94.7% to 0% for the sinusoidal model. This establishes an association between extrapolation failure and a redistribution of attention in these checkpoints. It does not, however, establish that increasing entropy caused the errors. The position-wise and carry-condition results provide more direct evidence about how Addition predictions failed.

5.5.2 Delayed Copy

Delayed Copy produced a different pattern. Normalised entropy remained approximately stable for both eligible encodings. For the sinusoidal model, the values were 0.645, 0.627, and 0.630 at lengths 128, 256, and 1024, respectively. For RoPE, they were 0.605, 0.577, and 0.638. Exact-match accuracy nevertheless fell from 99.8% and 100% at the training length to 0% at length 256.

The absence of a substantial entropy increase at the first unseen length is important. It shows that immediate extrapolation collapse was not always accompanied by greater attention diffusion. The local attention ratio decreased for both encodings, but this statistic uses a fixed ± 10 -position neighbourhood. Its value is therefore expected to decrease partly as the total sequence length increases and must be interpreted cautiously.

5.5.3 Reverse

For Reverse with learned positional embeddings, normalised entropy increased from 0.539 at length 128 to 0.731 at length 256 and 0.910 at length 1024. The local attention ratio simultaneously decreased from 0.119 to 0.073 and then 0.022. The first change coincided with exact-match accuracy falling from 100% to 0%.

These results show that the representative Reverse model’s attention became more diffuse outside the training length. However, the position-wise and dependency-distance analyses show whether the resulting prediction errors were spatially concentrated. The attention measurements alone cannot determine whether the model learned a length-specific reversal mechanism.

5.5.4 Interpretation Across Tasks

Average attention distance increased with test length in every task. However, this metric is measured in raw token positions, so part of the increase follows mechanically from the larger set of possible distances in longer sequences. It is therefore not treated as independent evidence of improved long-range reasoning or of failure.

Taken together, the attention diagnostics do not support a single universal attention-based explanation. Addition and Reverse showed increased normalised entropy, whereas Delayed Copy collapsed without a comparable increase at the first unseen length. Attention change is therefore retained as descriptive supporting evidence rather than used as a separate failure category in the final taxonomy.

5.6 Mitigation Strategy Design

The mitigation experiments test specific hypotheses about why a model that learns the training task might fail at longer lengths. The purpose is not simply to train additional variants, but to determine whether changing the distribution or ordering of training lengths alters the observed extrapolation behaviour.

The experiments use representative seed-42 configurations. A mitigation is considered interpretable as a length-generalisation test only when it first preserves training-length performance. The same 90% exact-match eligibility threshold used for the baseline analysis is therefore applied to the mitigation variants. If a variant falls below this threshold, its out-of-distribution results are classified as inconclusive because the training task was not learned sufficiently.

Condition	Applied setting	Specific question tested
Single-length control	Delayed Copy, sinusoidal	Does the mitigation training pipeline reproduce the original result when the length distribution remains unchanged?
Mixed-length training	Addition, learned; Delayed Copy, sinusoidal	Does simultaneous exposure to several lengths reduce specialisation to one fixed training length? Addition uses lengths 4, 8, 12, and 16; Delayed Copy uses lengths 32, 64, 96, and 128.
Strengthened mixed-length training	Addition, learned	Was the initial mixed-length result caused by insufficient data or optimisation time? The second variant increases both the number of training examples and the number of epochs.
Curriculum training	Delayed Copy, sinusoidal and RoPE; Reverse, learned	Does sequential training at lengths 32, 64, 96, and 128 reduce an optimisation barrier associated with training directly at length 128?
Randomised padded training	Delayed Copy, sinusoidal and RoPE	Does combining variable-length examples in a shuffled, right-padded dataset reduce reliance on fixed separator and output positions? This intervention varies effective sequence length but does not randomly shift tokens to new absolute offsets.

Table 5.2: Mitigation conditions and the specific hypotheses tested by each intervention.

The primary comparison is exact-match accuracy at the training length and the first evaluated unseen length. The longest test length is also reported, but it is not used alone to judge mitigation success. A useful mitigation would preserve training-length performance and produce a meaningful improvement over the corresponding baseline at unseen lengths.

5.7 Mitigation Results

The mitigation experiments produced two principal outcomes. Some variants failed to learn the training task and therefore did not provide a valid test of extrapolation. Other variants preserved training-length performance but showed no observed improvement at the first unseen or longest evaluated length.

These experiments use one representative seed and should consequently be interpreted as targeted intervention tests rather than multi-seed estimates.

5.7.1 Addition: Mixed-Length Training

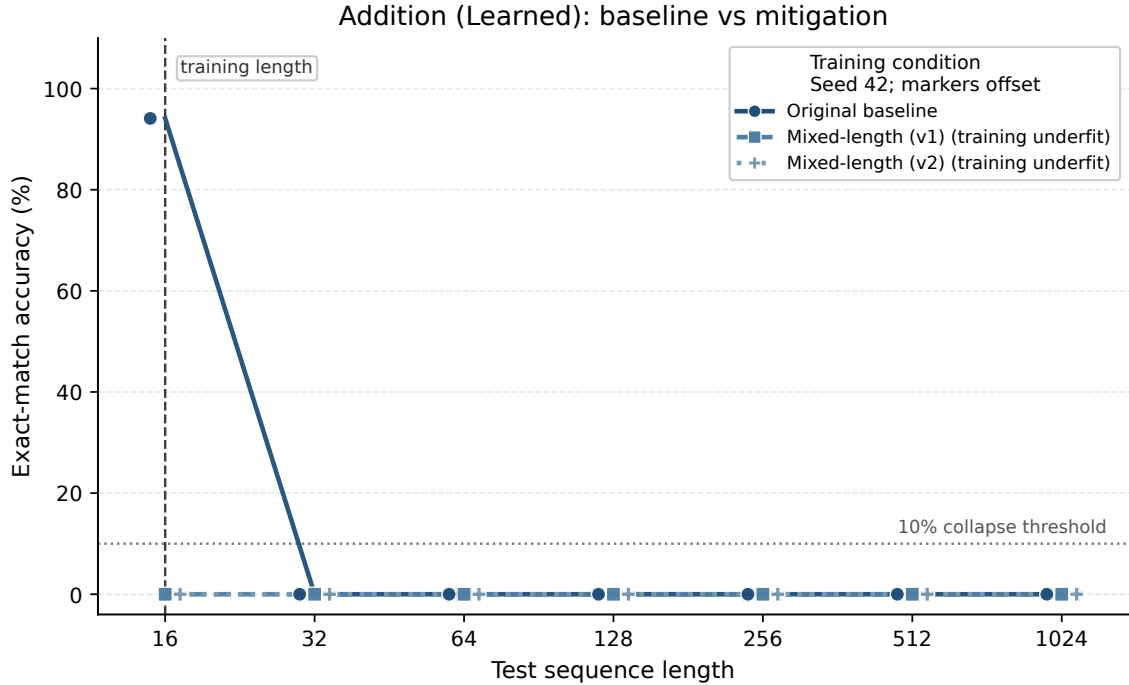


Figure 5.6: Exact-match accuracy for the original Addition learned baseline and two mixed-length variants. Results use seed 42. Both mixed-length variants fall below the training-length eligibility threshold, so their extrapolation results are classified as inconclusive.

Figure 5.6 compares the original learned-position baseline with two mixed-length variants. The original baseline achieved 94.1% exact-match accuracy at length 16 and 0% at the first unseen length of 32.

The first mixed-length variant achieved 0% exact-match accuracy at length 16. The strengthened variant, which used more examples and additional epochs, also achieved 0% at the training length. Because neither variant learned the length-16 task, their 0% performance at longer lengths cannot be interpreted as evidence about extrapolation.

The appropriate conclusion is therefore not that mixed-length training was ineffective at improving an otherwise learned model. Instead, mixed-length training introduced a training-length optimisation failure under both evaluated settings. Whether a differently tuned mixed-length procedure could preserve the baseline fit remains unresolved.

5.7.2 Delayed Copy with Sinusoidal Encoding

The single-length control reproduced the original sinusoidal baseline exactly, with a maximum difference of 0 percentage points across the evaluated lengths. Both achieved 99.8% exact-match accuracy at length 128 and 0% at lengths 256, 512, and 1024. This confirms that the mitigation pipeline could reproduce the reference failure curve when the training-length distribution was unchanged.

The mixed-length variant achieved 0% exact-match accuracy at the training length and is therefore classified as training-length underfitting. In contrast, curriculum and randomised-padded training both achieved 100% exact-match accuracy at length 128. These two variants provide valid extrapolation tests because they preserved training-length performance.

Neither valid mitigation improved performance at the first unseen length. Curriculum and randomised-padded training both achieved 0% exact-match accuracy at lengths 256, 512, and 1024, matching the extrapolation behaviour of the original baseline. Thus, gradual length exposure and jointly padded variable-length exposure produced no observed exact-match improvement in this configuration.

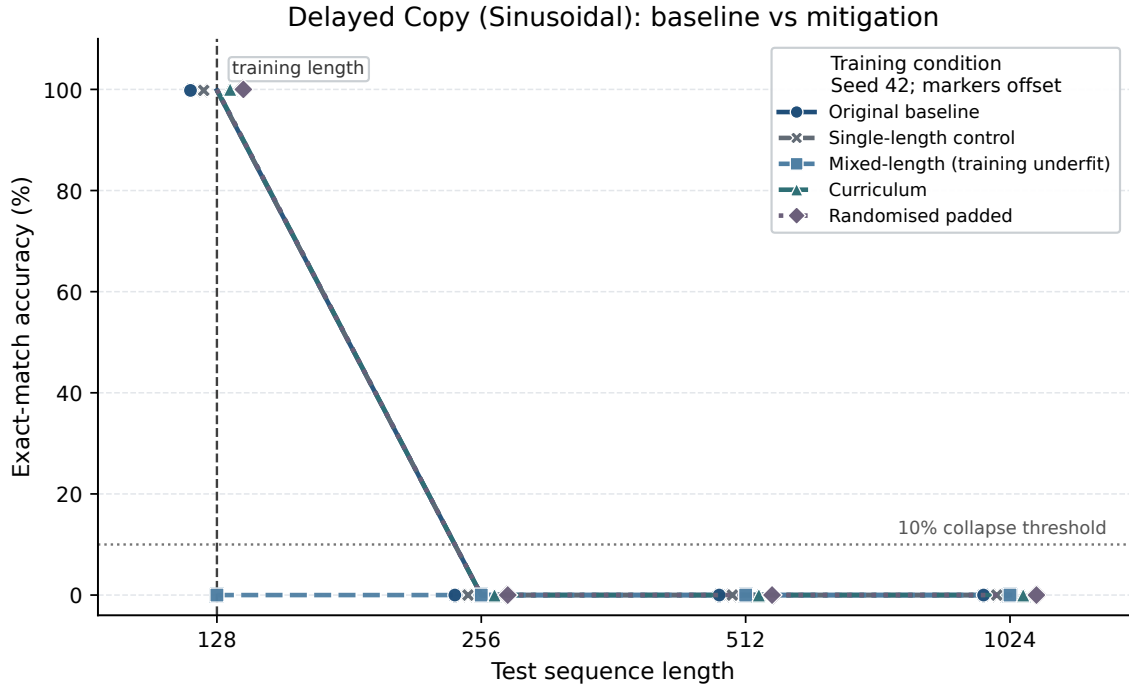


Figure 5.7: Exact-match accuracy for the original Delayed Copy sinusoidal baseline, the single-length control, and the mixed-length, curriculum, and randomised-padded variants. Results use seed 42; no multi-seed error bars are shown.

5.7.3 Delayed Copy with RoPE

Figure 5.8 shows that the original baseline, curriculum variant, and randomised-padded variant all achieved 100% exact-match accuracy at length 128. Training-length underfitting therefore does not explain the mitigation outcomes for this configuration.

All three conditions nevertheless achieved 0% exact-match accuracy at lengths 256, 512, and 1024. The two interventions preserved the ability to solve the training task but produced no observed improvement at unseen lengths. This result is consistent with the sinusoidal experiment and shows that the outcome was not limited to one positional encoding.

5.7.4 Reverse Curriculum Qualification

The Reverse curriculum run achieved 100% exact-match accuracy at length 128 and 0% at the evaluated unseen lengths. However, this comparison is not fully matched: the original Reverse baseline used a vocabulary size of 100, whereas the curriculum configuration used a vocabulary size of 102. Consequently, the run is treated as exploratory supporting evidence rather than as a controlled baseline-mitigation comparison. A corrected rerun would be required before making a stronger claim about curriculum training on Reverse.

5.7.5 Interpretation of the Mitigation Evidence

The mitigation results do not support the broad claim that exposure to varied training lengths necessarily improves extrapolation. Mixed-length variants underfit in the evaluated Addition and sinusoidal Delayed Copy settings. Curriculum and randomised-padded Delayed Copy variants retained training-length performance but showed no exact-match improvement at the first unseen length.

These are bounded negative results. They show that the specific interventions, configurations, and representative seed evaluated here did not improve exact-match extrapolation. They do not establish that the failures are universally resistant to mitigation or that more extensive hyperparameter tuning, alternative curricula, or different model architectures could not improve performance.

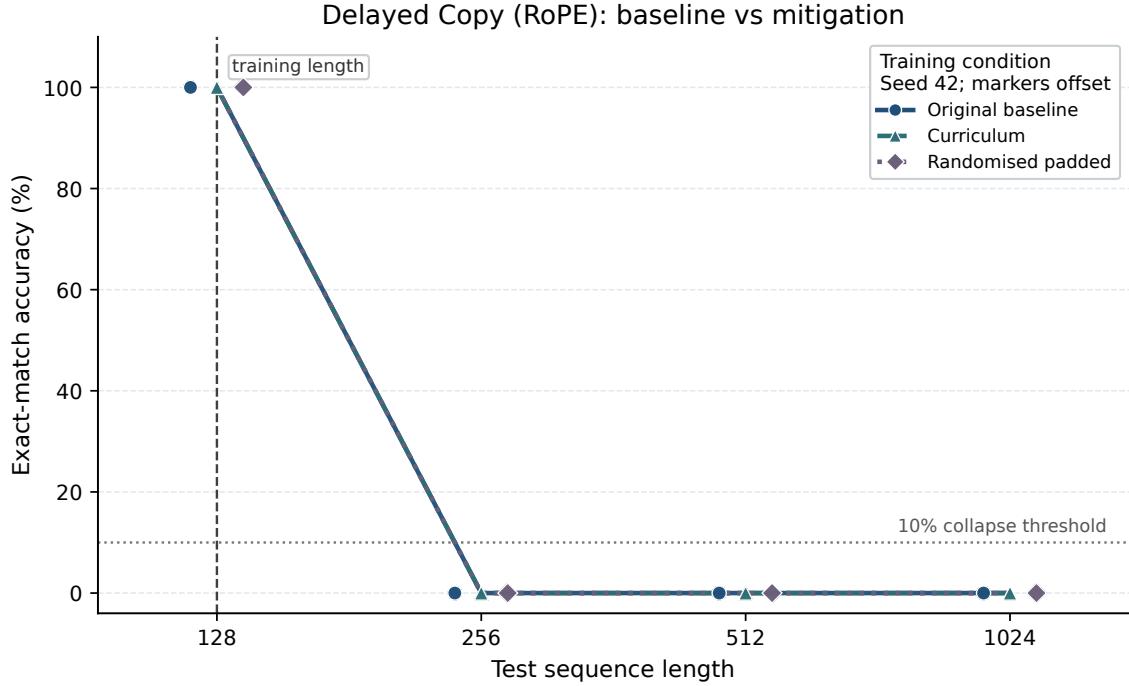


Figure 5.8: Exact-match accuracy for the original Delayed Copy RoPE baseline and the curriculum and randomised-padded variants. All conditions achieve 100% at the training length but 0% at every evaluated unseen length. Results use seed 42; no multi-seed error bars are shown.

5.8 Proposed Failure Taxonomy

The failure taxonomy is the main interpretive contribution of this project. Its purpose is to prevent distinct experimental outcomes from being grouped under the general statement that a model “fails at longer lengths”. The categories separate failure to learn the original task, instability across random seeds, failure after a valid training fit, the spatial form of the resulting errors, and the outcome of a tested intervention.

The taxonomy is applied sequentially. Training-length performance is checked first because a model must learn the original task before its extrapolation can be evaluated. Multi-seed consistency is then examined to determine whether the baseline is reliable. For eligible configurations, the first unseen length identifies whether immediate extrapolation collapse occurs. Position-wise and task-specific analyses subsequently characterise the spatial form of that collapse. Finally, mitigation outcomes are evaluated only when the intervention preserves training-length performance.

This ordering prevents several misleading comparisons. For example, NoPE and ALiBi frequently perform poorly in the present setup, but their outcomes are classified as training-length underfitting rather than extrapolation collapse. Delayed Copy with learned positional embeddings and Reverse with sinusoidal encoding are classified as seed-unstable because only some seeds learn the training task. In contrast, the five eligible task-encoding configurations provide valid evidence of immediate extrapolation collapse.

Attention behaviour is not used as a separate failure category. Addition and Reverse show increasing normalised attention entropy, but Delayed Copy collapses without a comparable entropy increase at the first unseen length. Attention statistics are therefore retained as descriptive supporting evidence rather than treated as a universal failure mechanism.

5.9 Summary

This chapter examined how the eligible models fail after leaving the training length. The position-wise analysis showed that Addition retains a modest error gradient, with lower errors near the beginning and near-reference errors towards later output positions. Delayed Copy and Reverse instead show almost uniform near-reference error across the output.

Failure category	Operational evidence	Interpretation
Training-length underfitting	Training-length exact-match accuracy is below 90%.	The task was not learned sufficiently, so longer-length performance cannot be used as evidence of extrapolation failure.
Seed instability	Only one or two of the three seeds reach the 90% training-fit threshold, or training performance varies substantially across seeds.	A pooled mean can conceal qualitatively different runs; successful and unsuccessful seeds should be reported separately.
Immediate extrapolation collapse	All three seeds satisfy baseline eligibility, but mean exact-match accuracy falls below 10% at the first evaluated unseen length.	Training-length success does not transfer to the first sampled longer length. “Immediate” refers to the evaluated length grid and does not locate the precise failure point between evaluations.
Spatial error structure	Position-wise or task-conditioned token-error rates vary across the target, carry condition, or dependency-distance bins.	This category describes how failure is distributed. Addition retains a modest positional gradient, whereas Delayed Copy and Reverse show almost uniform near-reference error.
No observed improvement under tested mitigation	A mitigation preserves training-length accuracy but does not improve exact-match accuracy at the first unseen length relative to the reference baseline.	The evaluated intervention did not improve extrapolation in the tested configuration. This is a scoped experimental result, not evidence of universal resistance to mitigation.

Table 5.3: Failure taxonomy derived from baseline eligibility, multi-seed robustness, position-wise diagnostics, and mitigation experiments.

The task-specific diagnostics further showed that Addition errors were not substantially concentrated on digits with an incoming carry and that Reverse errors did not increase meaningfully with dependency distance. Attention statistics produced task-dependent patterns: entropy increased for Addition and Reverse but remained approximately stable for Delayed Copy at the first unseen length. Attention diffusion therefore does not provide a universal explanation of the observed failures.

The mitigation experiments also produced distinct outcomes. Mixed-length variants underfit in the evaluated settings, while curriculum and randomised-padded Delayed Copy variants preserved training-length performance but produced no observed improvement at unseen lengths. These results were organised into five categories: training-length underfitting, seed instability, immediate extrapolation collapse, spatial error structure, and no observed improvement under tested mitigation.

The taxonomy should be interpreted as an evidence-based framework for the experiments in this dissertation rather than as an exhaustive taxonomy of all possible length-generalisation failures. It provides the basis for the broader interpretation, limitations, and recommendations developed in the following chapters.

Chapter 6

Exploratory Real-World Extension

The main experiments use synthetic algorithmic tasks because they provide controlled tests of length extrapolation and permit exact analysis of output errors. However, these tasks are deliberately simplified and do not represent all properties of applied sequential data. This chapter presents a limited real-world extension using the SelfRegulationSCP2 multivariate time-series classification dataset.

The purpose of this experiment is to provide an exploratory contrast rather than to validate the synthetic findings or establish state-of-the-art performance. In particular, classification accuracy and sequence-level exact-match accuracy measure different forms of performance and should not be compared directly.

6.1 Dataset and Task

SelfRegulationSCP2 is part of the UEA multivariate time-series classification archive [3]. It contains EEG recordings from a binary classification task involving the self-regulation of slow cortical potentials. The predefined split contains 200 training examples and 180 test examples. Each example contains seven channels and 1152 time points.

The model predicts one class label for each input sequence. This differs from the synthetic tasks, where a prediction is counted as an exact match only when every valid output token is correct. A sequence-level classifier may use partial temporal evidence, while the synthetic tasks require a complete structured output.

Component	Setting
Dataset	SelfRegulationSCP2
Data type	Multivariate EEG time series
Training examples	200
Test examples	180
Input channels	7
Classes	2
Full sequence length	1152
Training prefix length	256
Evaluation prefix lengths	256, 512, 1024, 1152
Metric	Classification accuracy
Positional encodings	Learned, sinusoidal, NoPE, ALiBi, RoPE
Seeds	42, 123, 2024

Table 6.1: Dataset and experimental configuration for the exploratory SelfRegulationSCP2 extension.

6.2 Experimental Protocol

The experiment follows a train-short/test-long prefix protocol. Each Transformer classifier is trained using the first 256 time points of every training example. The resulting checkpoint is evaluated on test prefixes of length 256, 512, 1024, and 1152. The test examples remain the same across lengths; only the amount of each time series supplied to the classifier changes.

Continuous input channels are projected from seven dimensions to the 128-dimensional Transformer representation using a linear layer. The model then applies the same two-block Transformer structure used in the synthetic experiments. Mean pooling across the temporal dimension produces a sequence-level representation, which is passed to a two-class linear classification head.

Normalisation statistics are calculated using only the training split and then applied to both training and test data. Each configuration is trained for 50 epochs using a batch size of 16 and repeated across three random seeds. Results are reported as mean test accuracy, with error bars denoting one standard error across the three seeds.

6.3 Real-World Results

The existing real-world plot should be regenerated in the final visual style before compilation. The revised figure will be saved as `selfregulationscp2_accuracy_mean_se.pdf`.

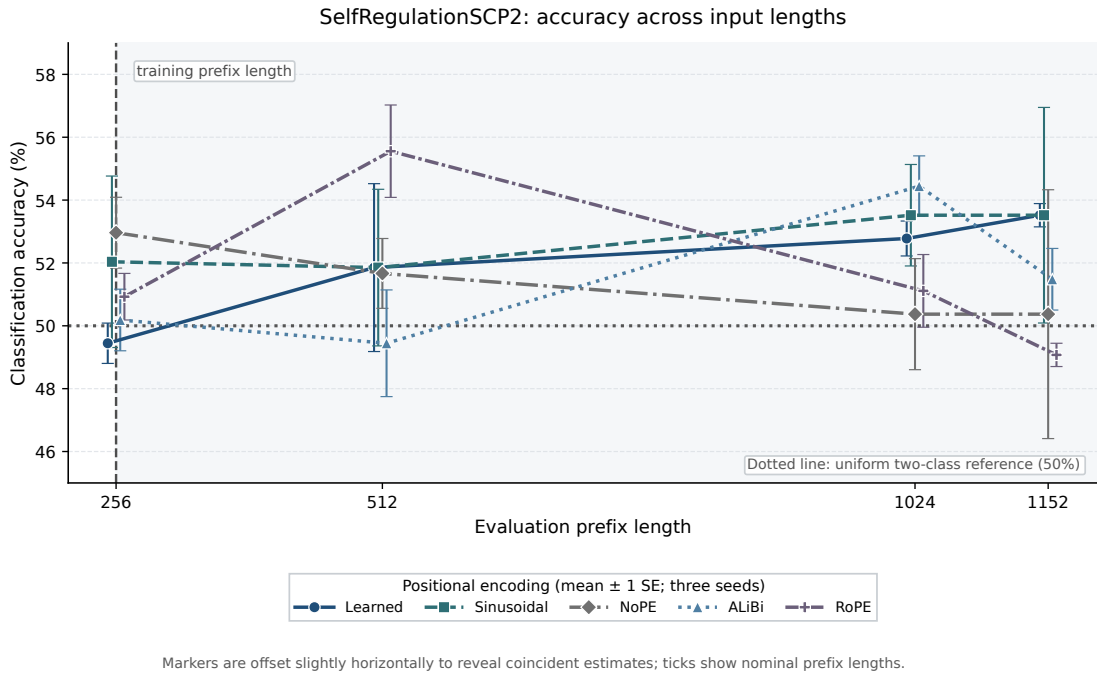


Figure 6.1: Mean SelfRegulationSCP2 test accuracy across three seeds as the evaluation prefix length increases. Error bars show ± 1 standard error. The dotted horizontal line denotes the 50% uniform two-class prediction reference, and the vertical marker identifies the training prefix length of 256. Markers are offset slightly horizontally to reveal coincident estimates; axis ticks show the nominal prefix lengths.

Encoding	Accuracy at 256 (%)	Accuracy at 1152 (%)	Change (percentage points)
Learned	49.44 ± 0.64	53.52 ± 0.37	+4.07
Sinusoidal	52.04 ± 2.73	53.52 ± 3.43	+1.48
NoPE	52.96 ± 1.13	50.37 ± 3.96	-2.59
ALiBi	50.19 ± 0.98	51.48 ± 0.98	+1.30
RoPE	50.93 ± 0.74	49.07 ± 0.37	-1.85

Table 6.2: Mean classification accuracy and standard error at the training prefix length and full sequence length. Changes are calculated from the unrounded mean accuracies.

Figure 6.1 shows that mean accuracy remains within approximately 49–56% across the evaluated lengths. No positional encoding exhibits the immediate transition from high training-length performance to zero accuracy observed in the eligible synthetic baselines.

At length 512, RoPE produces the highest individual mean of 55.56%, but its mean accuracy decreases to 49.07% at length 1152. At the full length, learned and sinusoidal encodings have the highest mean accuracy at 53.52%. However, these differences are small relative to the overall performance level and

seed variation, and no statistical significance test was conducted. Consequently, the experiment does not establish a reliable ranking among the positional encodings.

The most important qualification is that held-out accuracy at the training prefix length is already modest. Mean accuracy at length 256 ranges only from 49.44% to 52.96%, close to the 50% uniform two-class reference. Therefore, the absence of a sharp decline cannot be interpreted as evidence that the models learned a robust classifier that generalises across length. Unlike the eligible synthetic baselines, the real-world experiment does not begin from strong held-out task performance.

6.4 Comparison with the Synthetic Benchmark

The real-world curve differs descriptively from the synthetic exact-match curves. Eligible synthetic configurations achieve at least 90% exact-match accuracy at the training length and then fall to 0% at the first evaluated unseen length. The SelfRegulationSCP2 models instead remain within a narrow, modest accuracy range as additional time points are supplied.

Several factors prevent a causal interpretation of this contrast. First, classification produces one label, whereas the synthetic tasks require an entire output sequence. Second, the evaluation metrics have substantially different sensitivities to partial correctness. Third, the SelfRegulationSCP2 models do not establish a strong held-out reference performance at length 256. Finally, longer prefixes introduce additional signal that may be informative, irrelevant, or noisy depending on the example.

The experiment therefore shows that catastrophic exact-match collapse is not visible in this particular classification curve. It does not show that the classifier learned a length-general computation, nor does it establish that real-world classification tasks are generally less affected by sequence length.

6.5 Limitations of the Extension

This extension has several important limitations. It uses only one relatively small dataset, with 200 training examples and 180 test examples. No specialised time-series baseline or state-of-the-art architecture is included, and the Transformer hyperparameters were selected primarily for comparability with the synthetic experiments rather than optimised for this dataset.

The models are evaluated only through classification accuracy. No position-wise output diagnostic is possible because the task produces one label per sequence. In addition, the three-seed results provide only a small estimate of variability, and no formal significance testing or confidence interval analysis was performed.

Most importantly, test accuracy at the training prefix length remains close to the uniform two-class reference. The experiment therefore cannot cleanly separate failure to generalise across examples from failure to generalise across input length. A stronger extension would first establish a competitive held-out baseline and would then examine whether that performance is preserved when the input-length distribution changes.

Future work should include additional long-sequence datasets, majority-class and specialised time-series baselines, task-specific hyperparameter selection, and a larger number of seeds.

6.6 Summary

The SelfRegulationSCP2 extension provides an exploratory contrast to the controlled synthetic benchmark. Mean classification accuracy remains within approximately 49–56% as the input prefix increases from 256 to 1152, rather than showing an immediate fall to zero.

However, performance is already close to the uniform two-class reference at the training prefix length. The experiment therefore provides evidence of a flat but weak classification curve, not successful real-world length generalisation. Its main value is to demonstrate why conclusions about length behaviour depend on baseline validity, task structure, output type, and evaluation metric.

Chapter 7

Discussion

This chapter interprets the experimental findings in relation to the primary research question and its five supporting questions. The central issue is not only whether performance decreases at longer lengths, but whether the observed outcome represents failure to learn the training task, instability across seeds, collapse after a valid training fit, a particular spatial error pattern, or no observed improvement under a tested intervention.

The results demonstrate that these distinctions materially change how poor long-length accuracy should be interpreted. They also show that neither a particular positional encoding nor exposure to varied training lengths is, by itself, sufficient to guarantee length-general behaviour in the evaluated configurations.

7.1 Answers to the Research Questions

7.1.1 RQ1: Which Configurations Reliably Learn the Training Task?

Five task-encoding configurations satisfied the baseline-eligibility criterion across all three seeds: Addition with learned and sinusoidal positional encodings, Delayed Copy with sinusoidal encoding and RoPE, and Reverse with learned positional embeddings. These configurations provide valid starting points for analysing extrapolation because they reliably learned the corresponding training-length task.

Other configurations produced qualitatively different outcomes. Delayed Copy with learned positional embeddings and Reverse with sinusoidal encoding succeeded for only some seeds. Pooling these successful and unsuccessful runs into a single mean would conceal the fact that the configurations were unstable at the training length. NoPE and ALiBi, together with several other task-encoding combinations, did not reliably learn the training task under the present architecture and optimisation settings.

This result supports the decision to treat baseline eligibility as a prerequisite for extrapolation analysis. More generally, reporting multiple runs is important for stochastic neural models because conclusions based on one seed may not represent the distribution of possible outcomes [16]. In this project, seed variation is not merely uncertainty around a common mean; in some configurations it separates successful task learning from complete training-length failure.

7.1.2 RQ2: When Does Eligible Performance Collapse?

All five eligible configurations fell to 0% exact-match accuracy at the first evaluated unseen length. Addition collapsed when the digit length increased from 16 to 32, while Delayed Copy and Reverse collapsed when the sequence length increased from 128 to 256. Performance remained at 0% exact match at all longer evaluated lengths.

The behaviour is therefore classified as immediate extrapolation collapse. However, “immediate” is defined relative to the sampled evaluation grid. The experiments establish that collapse occurred by lengths 32 and 256, respectively, but do not identify the precise transition point between the training length and the first evaluated unseen length.

This result is consistent with previous evidence that strong performance on short reasoning problems does not necessarily transfer to longer instances [1]. It is also compatible with the argument that Transformer length generalisation depends on whether optimisation discovers a computation whose structure remains valid as length changes [23].

The results do not prove that a particular internal algorithm was learned at the training length. They establish the more limited empirical conclusion that the behaviour producing high training-length accuracy did not preserve complete sequence-level correctness at the first sampled longer length.

7.1.3 RQ3: Where Within the Output Do Errors Occur?

The position-wise diagnostics show that 0% exact-match accuracy does not correspond to an identical token-error pattern across tasks. At the first unseen length, Addition showed a modest positional gradient. Errors were lower near the beginning of the output and increased towards later positions, where they approached the 90% uniform digit-prediction reference. The sinusoidal model showed a larger positional range than the learned model.

Delayed Copy and Reverse exhibited a different form of failure. Their token-error rates were close to the 99% uniform valid-token reference across almost every relative output position. These configurations changed from approximately uniform correctness at the training length to approximately uniform incorrectness at the first unseen length.

The task-specific diagnostics further narrow the interpretation. For Addition, ordinary digits with and without an incoming carry had almost identical error rates. At the first unseen length, their difference was less than one percentage point for the learned model and effectively zero for the sinusoidal model. The failure was therefore not concentrated specifically on carry-dependent digits.

For Reverse, error rates were similarly high across short and long normalised input–output dependency distances. The failure cannot therefore be described as a progressive inability to handle only the most distant reverse mappings. Once the model left the training length, mappings across the full dependency-distance range failed at comparable rates.

These findings are important because they replace a general statement that the models “fail” with a description of failure behaviour. Addition retains some position-dependent structure, whereas Delayed Copy and Reverse show distributed near-reference collapse.

7.1.4 RQ4: How Does Positional Encoding Affect the Outcome?

Positional encoding strongly affected whether the model learned the training-length task, but no encoding was uniformly successful across all three tasks. Learned positional embeddings were eligible for Addition and Reverse but seed-unstable for Delayed Copy. Sinusoidal encoding was eligible for Addition and Delayed Copy but seed-unstable for Reverse. RoPE was eligible for Delayed Copy but underfit or only partially fit the other tasks.

Consequently, the experiments do not support a universal ranking of positional encodings. Their effects depend on the task, architecture, attention mask, and optimisation procedure. This task dependence is consistent with broader evidence that positional encoding is an important but context-dependent factor in Transformer length generalisation [9].

The underfitting of NoPE and ALiBi should not be interpreted as a general failure of those methods. Previous studies have reported effective length-related behaviour for NoPE in decoder-only Transformers [9] and for ALiBi in causal language modelling [15]. The present model is instead a compact, encoder-style Transformer using bidirectional attention. Its ALiBi implementation applies a symmetric distance bias rather than the original causal formulation. These architectural differences prevent a direct method-level comparison.

Similarly, RoPE provides a systematic rotation of query and key representations [19], but this representational mechanism does not by itself guarantee that optimisation will discover a length-general task solution. RoPE fitted Delayed Copy consistently and still collapsed at the first unseen length.

7.1.5 RQ5: Do the Tested Mitigations Improve Extrapolation?

The mitigation experiments produced two distinct outcomes. The Addition mixed-length variants and the sinusoidal Delayed Copy mixed-length variant underfit at the training length. Their long-length results are therefore inconclusive as tests of extrapolation.

Curriculum and randomised-padded training preserved training-length performance for the eligible Delayed Copy configurations. Nevertheless, they produced 0% exact-match accuracy at the first unseen length, matching the original baselines. These constitute valid negative results: the interventions preserved the training task but produced no observed extrapolation improvement in the tested settings.

The curriculum outcome does not show that curriculum learning is generally ineffective. Curriculum learning is a broad training principle in which examples are presented in an intended order of difficulty [5]. The present experiment tests only one four-stage length curriculum and one representative seed.

The randomised-padded experiment must also be distinguished from randomised positional encoding. The intervention used here combines shuffled, variable-length examples in a right-padded dataset. It does not sample ordered positional indices from a larger range. It therefore does not implement the method proposed by Ruoss et al., who directly randomise positional indices to expose the model to a broader positional distribution [17]. The negative result reported here does not contradict their findings.

The evidence supports the scoped category “no observed improvement under tested mitigation”. It does not support the stronger claim that the failures are inherently resistant to mitigation.

7.2 Contribution of the Failure Taxonomy

The main interpretive contribution is a five-part taxonomy based on observable experimental evidence:

1. training-length underfitting;
2. seed instability;
3. immediate extrapolation collapse;
4. spatial error structure; and
5. no observed improvement under tested mitigation.

These categories are not intended to represent mutually exclusive causal mechanisms. Instead, they form a sequential framework for interpreting experimental outcomes.

Training-length performance is examined first because extrapolation cannot be meaningfully assessed before the original task is learned. Seed consistency is then checked because a high pooled mean may hide failed runs. Eligible configurations are evaluated across unseen lengths to identify the first observed collapse. Position-wise and task-specific diagnostics characterise how errors are distributed. Finally, an intervention is judged only after checking whether it preserves the training baseline.

This structure provides more information than a single long-length accuracy table. For example, an ALiBi configuration that never learns the training task and a learned-position configuration that moves from 100% to 0% exact match may have the same accuracy at the longest length, but they do not provide the same evidence about length extrapolation.

The taxonomy is deliberately empirical and scoped to this dissertation. It should not be interpreted as an exhaustive theoretical classification of all possible Transformer failures. Its value lies in making experimental claims more precise and in exposing which additional evidence is needed before a particular conclusion can be drawn.

7.3 Interpretation of Attention Evidence

Attention statistics produced task-dependent rather than universal behaviour. For Addition and Reverse, normalised attention entropy increased as the test length grew. Delayed Copy, however, collapsed at the first unseen length without a comparable entropy increase. Attention diffusion therefore cannot serve as a general explanation of the observed failures.

The local attention ratio decreased with length across the analysed models, but this statistic uses a fixed ± 10 -position neighbourhood. Part of the decrease follows mechanically from evaluating a fixed window inside a larger sequence. Average attention distance also increased with length, but it is measured in raw token positions and therefore grows partly because longer sequences permit larger distances.

These limitations illustrate why attention weights should not automatically be treated as explanations of model predictions. Prior work has shown that different attention distributions can sometimes produce similar outputs and that attention weights need not correspond to other measures of feature importance [8]. Accordingly, this dissertation treats attention statistics as descriptive supporting evidence rather than as a causal mechanism or a separate taxonomy category.

A stronger mechanistic study would require targeted interventions, such as ablating particular heads, replacing learned attention patterns, or measuring whether controlled changes in attention alter the model’s output errors.

7.4 Interpretation of the Real-World Extension

The SelfRegulationSCP2 experiment produced a flatter accuracy curve than the synthetic benchmark. Across the evaluated prefix lengths and positional encodings, mean classification accuracy remained within approximately 49–56%. It did not show the transition from high training-length accuracy to zero performance observed in the eligible synthetic baselines.

This contrast must be interpreted cautiously. SelfRegulationSCP2 is a two-class EEG time-series dataset with only 200 training examples and 180 test examples [3]. Held-out accuracy at the training prefix length was already close to the 50% uniform two-class reference. The absence of a subsequent collapse therefore does not demonstrate successful length-general classification.

The synthetic and real-world experiments also use different output structures and evaluation metrics. Exact match requires every valid output token to be correct, while time-series classification requires one class label. In addition, longer SelfRegulationSCP2 prefixes introduce additional observations that may be informative, irrelevant, or noisy.

The extension therefore demonstrates a flat but weak classification curve, not evidence that real-world tasks are generally protected from length-related failure. Its broader lesson is methodological: the validity of the starting baseline and the meaning of the evaluation metric must be established before comparing behaviour across lengths or task types.

7.5 Limitations and Threats to Validity

7.5.1 Internal Validity

The main synthetic benchmark uses three random seeds. This is stronger than reporting one run, but it remains a small sample for estimating a complete performance distribution. The position-wise, attention, and mitigation diagnostics use representative seed-42 checkpoints and therefore characterise selected failure cases rather than multi-seed prevalence.

The mitigation experiments are not exhaustive hyperparameter searches. Moreover, the original Delayed Copy and Reverse baselines used a different batch size from several mitigation runs. The sinusoidal Delayed Copy single-length control reproduced the original baseline curve, providing a useful check for that pipeline, but equivalent controls were not run for every task–encoding pair.

The Reverse curriculum comparison has an additional configuration mismatch: the original baseline used a vocabulary size of 100, whereas the curriculum configuration used 102. That experiment is therefore exploratory and cannot be treated as a fully controlled baseline–mitigation comparison.

7.5.2 Construct Validity

Exact-match accuracy is appropriate for deterministic algorithmic tasks, but it is deliberately strict. A single incorrect output token makes the entire example incorrect. Token-error percentages and position-wise diagnostics were included to prevent this strict metric from hiding partial or spatially structured behaviour.

The first failure length is observed only on a discrete grid. The experiments cannot determine whether collapse begins immediately above the training length or only close to the first sampled unseen length.

Attention statistics are summary measures rather than causal tests. They are averaged over a limited number of diagnostic batches and should not be interpreted as full mechanistic explanations.

The positional encoding implementations also differ from some original settings in the literature. In particular, the present Transformer is bidirectional, and ALiBi uses a symmetric distance bias. Learned positional entries beyond the training length exist in the model but receive no direct training signal. Conclusions therefore apply to the implementations tested here rather than every possible form of the named encoding.

7.5.3 External Validity

The synthetic experiments use one compact Transformer architecture and three deterministic tasks. Larger models, decoder-only architectures, causal attention, alternative optimisation procedures, or tasks requiring different computations may produce different results.

Algorithmic reasoning studies have shown that Transformer length generalisation can vary considerably with task representation, scratchpad format, and the complexity of the required computation

[1, 23]. The present negative results should therefore not be generalised to every Transformer or training procedure.

The real-world extension contains only one small dataset and no specialised time-series baseline. It provides contextual evidence rather than a broad empirical claim about applied long-sequence classification.

7.6 Implications and Future Work

The principal methodological implication is that length-generalisation evaluation should proceed in stages:

1. establish that the training task is reliably learned;
2. evaluate multiple random seeds before pooling results;
3. report performance across a sequence of unseen lengths;
4. examine token-level and position-wise error behaviour;
5. use task-specific diagnostics where possible; and
6. compare interventions with an appropriate reference baseline.

This procedure distinguishes the ability to process a long context from the ability to preserve a learned computation under a length shift. Architectures and benchmarks designed for long sequences demonstrate that longer contexts can be processed efficiently [6, 21], but context capacity alone does not establish algorithmic length generalisation.

Future experiments should evaluate more seeds and denser length grids, especially between the training length and the first observed collapse. Hyperparameters should be selected separately for baseline fitting and mitigation evaluation, with matched batch sizes and exact configuration checks.

The Reverse curriculum experiment should be repeated with vocabulary size 100 to provide a fully matched comparison. A direct implementation of randomised positional indices [17] would test a different hypothesis from right-padded variable-length training. Scratchpad or intermediate-step representations could also be investigated because their format has been shown to affect length generalisation in reasoning tasks [1, 9].

Finally, a stronger real-world extension should first establish competitive held-out performance at the training prefix length. It should then evaluate multiple datasets, task-specific time-series baselines, additional seeds, and input-length interventions designed around the information structure of each dataset.

7.7 Summary

The discussion shows that the experimental outcomes answer different questions and should not be reduced to a single measure of long-length accuracy. Five task-encoding configurations reliably learned their training tasks and then collapsed at the first evaluated unseen length. Other configurations underfit or were unstable across seeds. Position-wise diagnostics distinguished a modest Addition error gradient from the nearly uniform failure of Delayed Copy and Reverse.

Attention statistics provided descriptive but non-universal evidence. Mixed-length variants underfit in the evaluated settings, while valid curriculum and randomised-padded Delayed Copy variants produced no observed improvement beyond the training length. The real-world extension showed a flat but weak classification curve and therefore did not establish successful length generalisation.

Together, these findings support an evidence-based taxonomy comprising training-length underfitting, seed instability, immediate extrapolation collapse, spatial error structure, and no observed improvement under tested mitigation. The main contribution is both the observed failure behaviour and the disciplined sequence of checks used to interpret it.

Chapter 8

Conclusion

This dissertation investigated which failure behaviours occur when compact Transformer models trained on short sequences are evaluated on substantially longer sequences. The study used a controlled train-short/test-long framework covering three synthetic tasks, five positional encoding settings, three random seeds, position-wise and task-specific diagnostics, targeted mitigation experiments, and an exploratory real-world extension.

The principal conclusion is that high training-length accuracy is not sufficient evidence of length-general computation. All five task-encoding configurations that satisfied the multi-seed baseline-eligibility criterion fell to 0% exact-match accuracy at the first evaluated unseen length. However, the broader experiment also showed that poor long-length accuracy does not always represent the same phenomenon. Some configurations underfit at the training length, some were unstable across seeds, and others reliably learned the training task before collapsing under length extrapolation.

8.1 Principal Findings

The baseline analysis identified five eligible configurations: Addition with learned and sinusoidal positional encodings, Delayed Copy with sinusoidal encoding and RoPE, and Reverse with learned positional embeddings. Addition collapsed when the digit length increased from 16 to 32, while Delayed Copy and Reverse collapsed when the sequence length increased from 128 to 256. These outcomes were reported in Chapter 4.

No positional encoding was uniformly successful across the three tasks. Positional encoding affected whether the training task was learned, but training-length success did not guarantee extrapolation. NoPE and ALiBi frequently underfit under the evaluated architecture and optimisation settings, while learned, sinusoidal, and rotary encodings produced task-dependent combinations of eligibility, seed instability, and extrapolation collapse.

The error diagnostics in Chapter 5 showed that the form of collapse also differed across tasks. At the first unseen length, Addition produced approximately 86.9–88.0% token error and retained a modest positional gradient, with lower error near the beginning and near-reference error towards later output positions. Delayed Copy and Reverse produced approximately 98.8–99.0% token error that was almost uniform across the output.

Addition errors differed by less than one percentage point between ordinary digits with and without an incoming carry. Reverse errors were similarly high across short and long normalised dependency distances. Therefore, Addition collapse was not primarily concentrated on carry-dependent digits, and Reverse collapse was not confined to the longest input-output dependencies.

The attention diagnostics produced task-dependent results. Normalised attention entropy increased for Addition and Reverse but remained approximately stable for Delayed Copy at the first unseen length. Attention change was therefore retained as descriptive evidence rather than treated as a universal explanation or an independent failure category.

The mitigation experiments also produced two distinct outcomes. Mixed-length variants underfit in the evaluated Addition and sinusoidal Delayed Copy settings, making their extrapolation results inconclusive. Curriculum and randomised-padded Delayed Copy variants preserved training-length performance but produced no observed improvement at unseen lengths. These results are scoped negative findings about the tested configurations; they do not show that the failures are universally resistant to mitigation.

Finally, the exploratory SelfRegulationSCP2 experiment in Chapter 6 produced mean classification accuracy of approximately 49–56% across the evaluated prefix lengths. The curve was relatively flat, but performance was already close to the uniform two-class reference at the training prefix length. It therefore represented flat but weak classification performance rather than successful real-world length generalisation.

8.2 Contributions

The dissertation makes five main contributions:

1. It establishes a baseline-eligibility procedure that separates failure to learn the training task from failure under length extrapolation.
2. It provides a controlled multi-task and multi-seed comparison of learned, sinusoidal, NoPE, ALiBi, and RoPE positional encoding settings.
3. It extends aggregate accuracy analysis with token-error percentages, position-wise profiles, carry-conditioned errors, and dependency-distance diagnostics.
4. It evaluates mixed-length, curriculum, and randomised-padded training against corresponding reference baselines while explicitly checking whether each intervention preserves training-length performance.
5. It proposes an evidence-based taxonomy that distinguishes five observable outcomes:
 - (a) training-length underfitting;
 - (b) seed instability;
 - (c) immediate extrapolation collapse;
 - (d) spatial error structure; and
 - (e) no observed improvement under tested mitigation.

The taxonomy is not intended as an exhaustive theory of every possible Transformer failure. Its contribution is methodological: it provides a structured sequence of checks that makes length-generalisation claims more precise and prevents different experimental outcomes from being collapsed into one statement that a model “fails”.

8.3 Limitations and Future Work

The conclusions are limited to one compact Transformer architecture, three synthetic tasks, five positional encoding implementations, and three baseline seeds. Future experiments should evaluate larger and alternative architectures, additional algorithmic tasks, more seeds, and denser test lengths between the training length and the first observed collapse.

Mitigation experiments should use fully matched optimisation settings and should be repeated across multiple seeds. In particular, the Reverse curriculum experiment should be repeated using vocabulary size 100 to match the original baseline. The effects of batch size and other optimisation choices should also be controlled through single-length reference runs.

Future interventions could test genuinely randomised positional indices, alternative curricula, intermediate-step or scratchpad representations, and architectures with iterative or recurrent computation. These experiments would test different hypotheses from the right-padded variable-length training used in this dissertation.

A stronger mechanistic analysis should move beyond aggregate attention statistics. Controlled head or layer ablations, attention replacement, probing experiments, and causal interventions could test whether particular internal changes contribute directly to output errors.

Finally, the real-world extension should be expanded only after establishing a stronger held-out baseline at the training prefix length. Future work should include multiple long-sequence datasets, specialised time-series baselines, task-specific hyperparameter selection, and additional random seeds.

8.4 Final Conclusion

The answer to the primary research question is that Transformer length-generalisation failure appears as a set of empirically distinguishable behaviours rather than one uniform reduction in accuracy. A valid analysis must first establish whether the training task was learned, determine whether that result is stable across seeds, locate the first observed extrapolation collapse, examine how errors are distributed, and test interventions against an appropriate baseline.

Within the evaluated synthetic benchmark, every eligible configuration collapsed at the first sampled unseen length. Yet the position-wise, task-specific, seed-level, and mitigation evidence showed that the surrounding failure behaviour differed substantially across configurations. The central contribution of this dissertation is therefore not only the observation that length extrapolation fails, but a reproducible framework for determining what kind of experimental failure has occurred and what conclusions the available evidence can support.

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Appendix A

Supplementary Position-Wise Error Heatmaps

Chapter 5 compared position-wise error at the training length and the first evaluated unseen length. This appendix provides the corresponding heatmaps across every evaluated sequence length.

The heatmaps use representative seed-42 checkpoints. Target positions are divided into ten equal-width relative-position bins. The shared colour scale shows token-error rate from 0% to 100%. These figures characterise selected failure cases and should not be interpreted as multi-seed estimates.

A.1 Addition

A.2 Delayed Copy

A.3 Reverse

A.4 Interpretive Scope

The heatmaps reinforce the distinction identified in the main analysis. Addition retains some positional variation at the first unseen length, particularly near the beginning of the target. Delayed Copy and Reverse instead show near-uniform high token error across positions immediately outside the training length. At longer lengths, all three tasks approach their respective uniform-prediction error references.

No additional statistical claims are based on these appendix figures. Their purpose is to provide the complete position-by-length visualisation underlying the more focused comparisons presented in the main chapter.

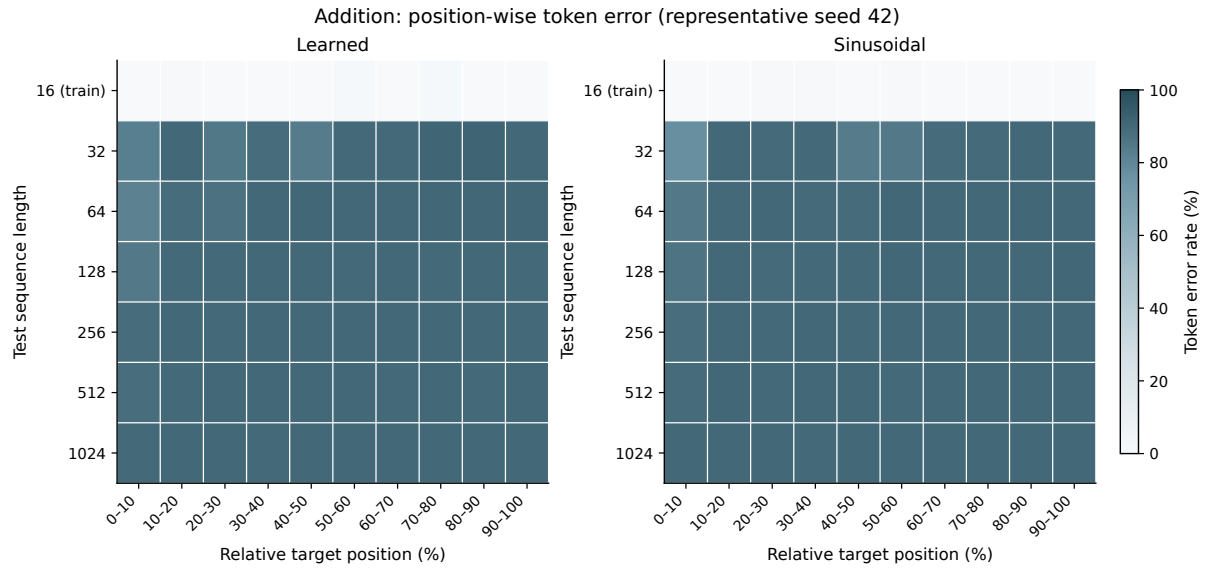


Figure A.1: Position-wise Addition token-error rates across all evaluated digit lengths for the learned and sinusoidal seed-42 checkpoints. Target positions are divided into ten relative-position bins, and all panels use the same 0–100% colour scale.

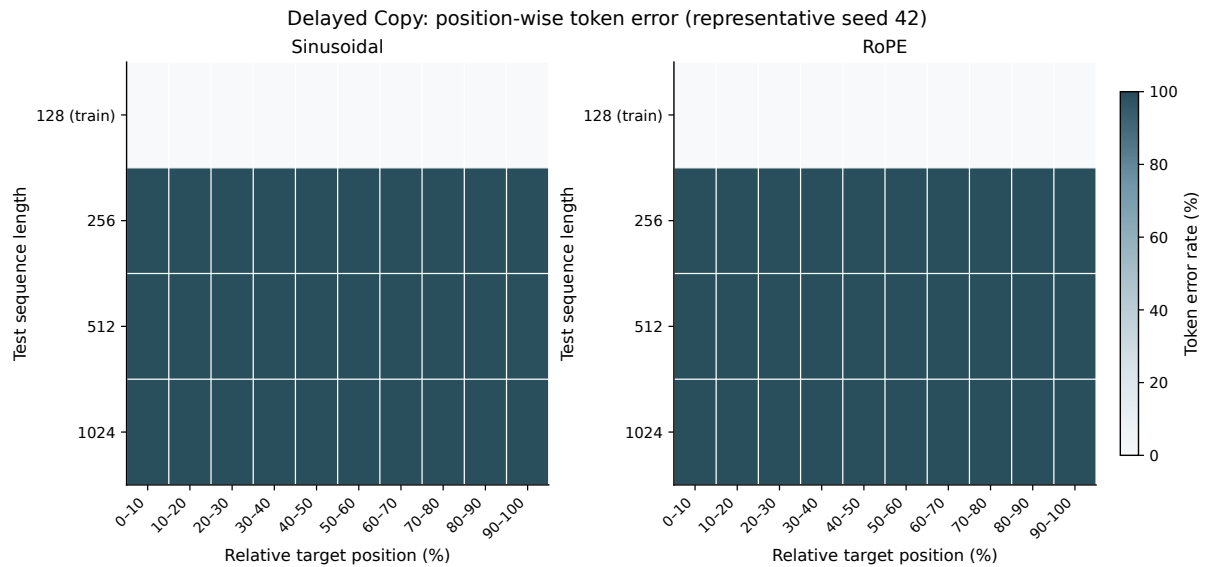


Figure A.2: Position-wise Delayed Copy token-error rates across all evaluated sequence lengths for the sinusoidal and RoPE seed-42 checkpoints. Target positions are divided into ten relative-position bins, and all panels use the same 0–100% colour scale.

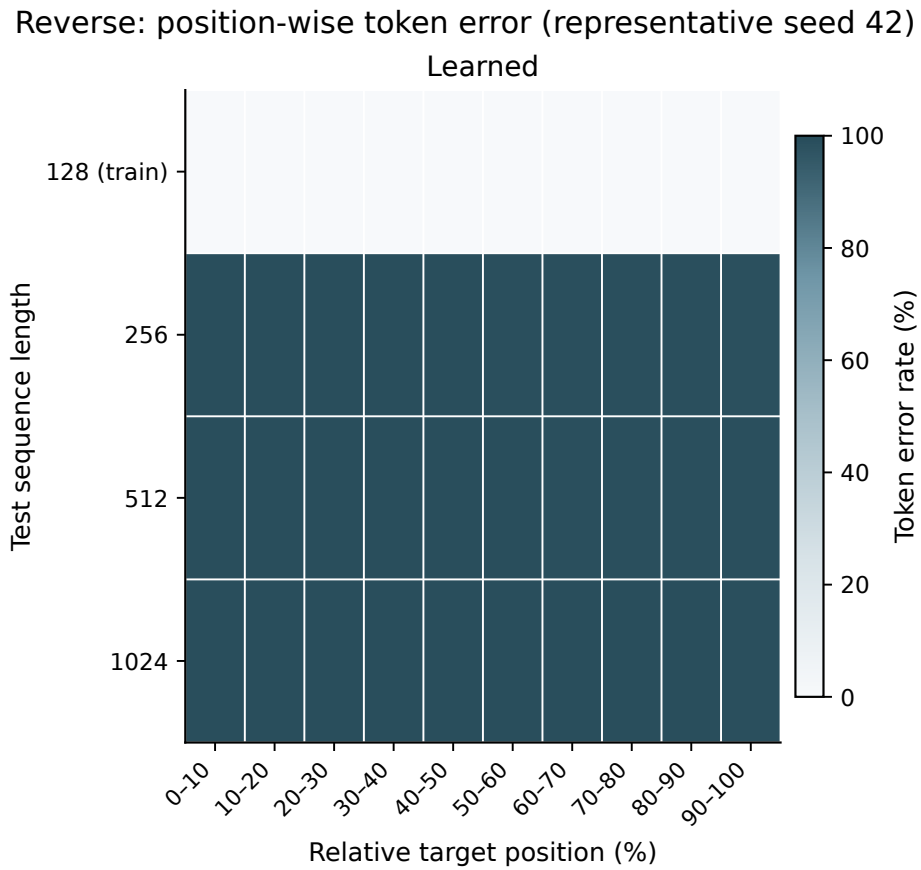


Figure A.3: Position-wise Reverse token-error rates across all evaluated sequence lengths for the learned seed-42 checkpoint. Target positions are divided into ten relative-position bins, and the colour scale shows token error from 0% to 100%.